

# SMOOTHING RING MAPS

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## 1. Introduction

The main result of this chapter is the following:

A regular map of Noetherian rings is a filtered colimit of smooth ones.

This theorem is due to Popescu, see [Pop90]. A readable exposition of Popescu's proof was given by Richard Swan, see [Swa98] who used notes by André and a paper of Ogoma, see [Ogo94].

Our exposition follows Swan's, but we first prove an intermediate result which lets us work in a slightly simpler situation. Here is an overview. We first solve the following “lifting problem”: A flat infinitesimal deformation of a filtered colimit of smooth algebras is a filtered colimit of smooth algebras. This result essentially says that it suffices to prove the main theorem for maps between reduced Noetherian rings. Next we prove two very clever lemmas called the “lifting lemma” and the “desingularization lemma”. We show that these lemmas combined reduce the main theorem to proving a Noetherian, geometrically regular algebra  $\Lambda$  over a field  $k$  is a filtered colimit of smooth  $k$ -algebras. Next, we discuss the necessary local tricks that go into the Popescu-Ogoma-Swan-André proof. Finally, in the last three sections we give the proof.

We end this introduction with some pointers to references. Let  $A$  be a henselian Noetherian local ring. We say  $A$  has the *approximation property* if for any  $f_1, \dots, f_m \in$

$A[x_1, \dots, x_n]$  the system of equations  $f_1 = 0, \dots, f_m = 0$  has a solution in the completion of  $A$  if and only if it has a solution in  $A$ . This definition is due to Artin. Artin first proved the approximation property for analytic systems of equations, see [Art68]. In [Art69] Artin proved the approximation property for local rings essentially of finite type over an excellent discrete valuation ring. Artin conjectured (page 26 of [Art69]) that every excellent henselian local ring should have the approximation property.

At some point in time it became a conjecture that every regular homomorphism of Noetherian rings is a filtered colimit of smooth algebras (see for example [Ray72], [Pop81], [Art82], [AD83]). We're not sure who this conjecture<sup>1</sup> is due to. The relationship with the approximation property is that if  $A \rightarrow A^\wedge$  is a colimit of smooth algebras with  $A$  as above, then the approximation property holds (insert future reference here). Moreover, the main theorem applies to the map  $A \rightarrow A^\wedge$  if  $A$  is an excellent local ring, as one of the conditions of an excellent local ring is that the formal fibres are geometrically regular. Note that excellent local rings were defined by Grothendieck and their definition appeared in print in 1965.

In [Art82] it was shown that  $R \rightarrow R^\wedge$  is a filtered colimit of smooth algebras for any local ring  $R$  essentially of finite type over a field. In [AR88] it was shown that  $R \rightarrow R^\wedge$  is a filtered colimit of smooth algebras for any local ring  $R$  essentially of finite type over an excellent discrete valuation ring. Finally, the main theorem was shown in [Pop85], [Pop86], [Pop90], [Ogo94], and [Swa98] as discussed above.

Conversely, using some of the results above, in [Rot90] it was shown that any Noetherian local ring with the approximation property is excellent.

The paper [Spi99] provides an alternative approach to the main theorem, but it seems hard to read (for example [Spi99, Lemma 5.2] appears to be an incorrectly reformulated version of [Elk73, Lemma 3]). There is also a Bourbaki lecture about this material, see [Tei95].

## 2. Singular ideals

Let  $R \rightarrow A$  be a ring map. The singular ideal of  $A$  over  $R$  is the radical ideal in  $A$  cutting out the singular locus of the morphism  $\text{Spec}(A) \rightarrow \text{Spec}(R)$ . Here is a formal definition.

**Definition 2.1.** Let  $R \rightarrow A$  be a ring map. The *singular ideal of  $A$  over  $R$* , denoted  $H_{A/R}$  is the unique radical ideal  $H_{A/R} \subset A$  with

$$V(H_{A/R}) = \{\mathfrak{q} \in \text{Spec}(A) \mid R \rightarrow A \text{ not smooth at } \mathfrak{q}\}$$

This makes sense because the set of primes where  $R \rightarrow A$  is smooth is open, see Algebra, Definition ???. In order to find an explicit set of generators for the singular ideal we first prove the following lemma.

**Lemma 2.2.** *Let  $R$  be a ring. Let  $A = R[x_1, \dots, x_n]/(f_1, \dots, f_m)$ . Let  $\mathfrak{q} \subset A$  be a prime ideal. Assume  $R \rightarrow A$  is smooth at  $\mathfrak{q}$ . Then there exists an  $a \in A$ ,  $a \notin \mathfrak{q}$ , an integer  $c$ ,  $0 \leq c \leq \min(n, m)$ , subsets  $U \subset \{1, \dots, n\}$ ,  $V \subset \{1, \dots, m\}$  of cardinality  $c$  such that*

$$a = a' \det(\partial f_j / \partial x_i)_{j \in V, i \in U}$$

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<sup>1</sup>The question/conjecture as formulated in [Art82], [AD83], and [Pop81] is stronger and was shown to be equivalent to the original version in [CP84].

for some  $a' \in A$  and

$$af_\ell \in (f_j, j \in V) + (f_1, \dots, f_m)^2$$

for all  $\ell \in \{1, \dots, m\}$ .

**Proof.** Set  $I = (f_1, \dots, f_m)$  so that the naive cotangent complex of  $A$  over  $R$  is homotopy equivalent to  $I/I^2 \rightarrow \bigoplus \text{Ad}x_i$ , see Algebra, Lemma ???. We will use the formation of the naive cotangent complex commutes with localization, see Algebra, Section ??, especially Algebra, Lemma ??. By Algebra, Definitions ?? and ?? we see that  $(I/I^2)_a \rightarrow \bigoplus A_a \text{d}x_i$  is a split injection for some  $a \in A$ ,  $a \notin \mathfrak{q}$ . After renumbering  $x_1, \dots, x_n$  and  $f_1, \dots, f_m$  we may assume that  $f_1, \dots, f_c$  form a basis for the vector space  $I/I^2 \otimes_A \kappa(\mathfrak{q})$  and that  $\text{d}x_{c+1}, \dots, \text{d}x_n$  map to a basis of  $\Omega_{A/R} \otimes_A \kappa(\mathfrak{q})$ . Hence after replacing  $a$  by  $aa'$  for some  $a' \in A$ ,  $a' \notin \mathfrak{q}$  we may assume  $f_1, \dots, f_c$  form a basis for  $(I/I^2)_a$  and that  $\text{d}x_{c+1}, \dots, \text{d}x_n$  map to a basis of  $(\Omega_{A/R})_a$ . In this situation  $a^N$  for some large integer  $N$  satisfies the conditions of the lemma (with  $U = V = \{1, \dots, c\}$ ).  $\square$

We will use the notion of a *strictly standard* element in  $A$  over  $R$ . Our notion is slightly weaker than the one in Swan's paper [Swa98]. We also define an *elementary standard* element to be one of the type we found in the lemma above. We compare the different types of elements in Lemma 3.7.

**Definition 2.3.** Let  $R \rightarrow A$  be a ring map of finite presentation. We say an element  $a \in A$  is *elementary standard in  $A$  over  $R$*  if there exists a presentation  $A = R[x_1, \dots, x_n]/(f_1, \dots, f_m)$  and  $0 \leq c \leq \min(n, m)$  such that

$$(2.3.1) \quad a = a' \det(\partial f_j / \partial x_i)_{i,j=1, \dots, c}$$

for some  $a' \in A$  and

$$(2.3.2) \quad af_{c+j} \in (f_1, \dots, f_c) + (f_1, \dots, f_m)^2$$

for  $j = 1, \dots, m - c$ . We say  $a \in A$  is *strictly standard in  $A$  over  $R$*  if there exists a presentation  $A = R[x_1, \dots, x_n]/(f_1, \dots, f_m)$  and  $0 \leq c \leq \min(n, m)$  such that

$$(2.3.3) \quad a = \sum_{I \subset \{1, \dots, n\}, |I|=c} a_I \det(\partial f_j / \partial x_i)_{j=1, \dots, c, i \in I}$$

for some  $a_I \in A$  and

$$(2.3.4) \quad af_{c+j} \in (f_1, \dots, f_c) + (f_1, \dots, f_m)^2$$

for  $j = 1, \dots, m - c$ .

The following lemma is useful to find implications of (2.3.3).

**Lemma 2.4.** Let  $R$  be a ring. Let  $A = R[x_1, \dots, x_n]/(f_1, \dots, f_m)$  and write  $I = (f_1, \dots, f_m)$ . Let  $a \in A$ . Then (2.3.3) implies there exists an  $A$ -linear map  $\psi : \bigoplus_{i=1, \dots, n} \text{Ad}x_i \rightarrow A^{\oplus c}$  such that the composition

$$A^{\oplus c} \xrightarrow{(f_1, \dots, f_c)} I/I^2 \xrightarrow{f \mapsto df} \bigoplus_{i=1, \dots, n} \text{Ad}x_i \xrightarrow{\psi} A^{\oplus c}$$

is multiplication by  $a$ . Conversely, if such a  $\psi$  exists, then  $a^c$  satisfies (2.3.3).

**Proof.** This is a special case of Algebra, Lemma ??.  $\square$

**Lemma 2.5** (Elkik). *Let  $R \rightarrow A$  be a ring map of finite presentation. The singular ideal  $H_{A/R}$  is the radical of the ideal generated by strictly standard elements in  $A$  over  $R$  and also the radical of the ideal generated by elementary standard elements in  $A$  over  $R$ .*

**Proof.** Assume  $a$  is strictly standard in  $A$  over  $R$ . We claim that  $A_a$  is smooth over  $R$ , which proves that  $a \in H_{A/R}$ . Namely, let  $A = R[x_1, \dots, x_n]/(f_1, \dots, f_m)$ ,  $c$ , and  $a' \in A$  be as in Definition 2.3. Write  $I = (f_1, \dots, f_m)$  so that the naive cotangent complex of  $A$  over  $R$  is given by  $I/I^2 \rightarrow \bigoplus \text{Ad}x_i$ . Assumption (2.3.4) implies that  $(I/I^2)_a$  is generated by the classes of  $f_1, \dots, f_c$ . Assumption (2.3.3) implies that the differential  $(I/I^2)_a \rightarrow \bigoplus A_a dx_i$  has a left inverse, see Lemma 2.4. Hence  $R \rightarrow A_a$  is smooth by definition and Algebra, Lemma ??.

Let  $H_e, H_s \subset A$  be the radical of the ideal generated by elementary, resp. strictly standard elements of  $A$  over  $R$ . By definition and what we just proved we have  $H_e \subset H_s \subset H_{A/R}$ . The inclusion  $H_{A/R} \subset H_e$  follows from Lemma 2.2.  $\square$

**Example 2.6.** The set of points where a finitely presented ring map is smooth needn't be a quasi-compact open. For example, let  $R = k[x, y_1, y_2, y_3, \dots]/(xy_i)$  and  $A = R/(x)$ . Then the smooth locus of  $R \rightarrow A$  is  $\bigcup D(y_i)$  which is not quasi-compact.

**Lemma 2.7.** *Let  $R \rightarrow A$  be a ring map of finite presentation. Let  $R \rightarrow R'$  be a ring map. If  $a \in A$  is elementary, resp. strictly standard in  $A$  over  $R$ , then  $a \otimes 1$  is elementary, resp. strictly standard in  $A \otimes_R R'$  over  $R'$ .*

**Proof.** If  $A = R[x_1, \dots, x_n]/(f_1, \dots, f_m)$  is a presentation of  $A$  over  $R$ , then  $A \otimes_R R' = R'[x_1, \dots, x_n]/(f'_1, \dots, f'_m)$  is a presentation of  $A \otimes_R R'$  over  $R'$ . Here  $f'_j$  is the image of  $f_j$  in  $R'[x_1, \dots, x_n]$ . Hence the result follows from the definitions.  $\square$

**Lemma 2.8.** *Let  $R \rightarrow A \rightarrow \Lambda$  be ring maps with  $A$  of finite presentation over  $R$ . Assume that  $H_{A/R}\Lambda = \Lambda$ . Then there exists a factorization  $A \rightarrow B \rightarrow \Lambda$  with  $B$  smooth over  $R$ .*

**Proof.** Choose  $f_1, \dots, f_r \in H_{A/R}$  and  $\lambda_1, \dots, \lambda_r \in \Lambda$  such that  $\sum f_i \lambda_i = 1$  in  $\Lambda$ . Set  $B = A[x_1, \dots, x_r]/(f_1 x_1 + \dots + f_r x_r - 1)$  and define  $B \rightarrow \Lambda$  by mapping  $x_i$  to  $\lambda_i$ . To check that  $B$  is smooth over  $R$  use that  $A_{f_i}$  is smooth over  $R$  by definition of  $H_{A/R}$  and that  $B_{f_i}$  is smooth over  $A_{f_i}$ . Details omitted.  $\square$

### 3. Presentations of algebras

Some of the results in this section are due to Elkik. Note that the algebra  $C$  in the following lemma is a symmetric algebra over  $A$ . Moreover, if  $R$  is Noetherian, then  $C$  is of finite presentation over  $R$ .

**Lemma 3.1.** *Let  $R$  be a ring and let  $A$  be a finitely presented  $R$ -algebra. There exists finite type  $R$ -algebra map  $A \rightarrow C$  which has a retraction with the following two properties*

- (1) *for each  $a \in A$  such that  $R \rightarrow A_a$  is a local complete intersection (More on Algebra, Definition ??) the ring  $C_a$  is smooth over  $A_a$  and has a presentation  $C_a = R[y_1, \dots, y_m]/J$  such that  $J/J^2$  is free over  $C_a$ , and*
- (2) *for each  $a \in A$  such that  $A_a$  is smooth over  $R$  the module  $\Omega_{C_a/R}$  is free over  $C_a$ .*

**Proof.** Choose a presentation  $A = R[x_1, \dots, x_n]/I$  and write  $I = (f_1, \dots, f_m)$ . Define the  $A$ -module  $K$  by the short exact sequence

$$0 \rightarrow K \rightarrow A^{\oplus m} \rightarrow I/I^2 \rightarrow 0$$

where the  $j$ th basis vector  $e_j$  in the middle is mapped to the class of  $f_j$  on the right. Set

$$C = \text{Sym}_A^*(I/I^2).$$

The retraction is just the projection onto the degree 0 part of  $C$ . We have a surjection  $R[x_1, \dots, x_n, y_1, \dots, y_m] \rightarrow C$  which maps  $y_j$  to the class of  $f_j$  in  $I/I^2$ . The kernel  $J$  of this map is generated by the elements  $f_1, \dots, f_m$  and by elements  $\sum h_j y_j$  with  $h_j \in R[x_1, \dots, x_n]$  such that  $\sum h_j e_j$  defines an element of  $K$ . By Algebra, Lemma ?? applied to  $R \rightarrow A \rightarrow C$  and the presentations above and More on Algebra, Lemma ?? there is a short exact sequence

$$(3.1.1) \quad I/I^2 \otimes_A C \rightarrow J/J^2 \rightarrow K \otimes_A C \rightarrow 0$$

of  $C$ -modules. Let  $h \in R[x_1, \dots, x_n]$  be an element with image  $a \in A$ . We will use as presentations for the localized rings

$$A_a = R[x_0, x_1, \dots, x_n]/I' \quad \text{and} \quad C_a = R[x_0, x_1, \dots, x_n, y_1, \dots, y_m]/J'$$

where  $I' = (hx_0 - 1, I)$  and  $J' = (hx_0 - 1, J)$ . Hence  $I'/(I')^2 = A_a \oplus (I/I^2)_a$  as  $A_a$ -modules and  $J'/(J')^2 = C_a \oplus (J/J^2)_a$  as  $C_a$ -modules. Thus we obtain

$$(3.1.2) \quad C_a \oplus I/I^2 \otimes_A C_a \rightarrow C_a \oplus (J/J^2)_a \rightarrow K \otimes_A C_a \rightarrow 0$$

as the sequence of Algebra, Lemma ?? corresponding to  $R \rightarrow A_a \rightarrow C_a$  and the presentations above.

Next, assume that  $a \in A$  is such that  $A_a$  is a local complete intersection over  $R$ . Then  $(I/I^2)_a$  is finite projective over  $A_a$ , see More on Algebra, Lemma ?. Hence we see  $K_a \oplus (I/I^2)_a \cong A_a^{\oplus m}$  is free. In particular  $K_a$  is finite projective too. By More on Algebra, Lemma ?? the sequence (3.1.2) is exact on the left. Hence

$$J'/(J')^2 \cong C_a \oplus I/I^2 \otimes_A C_a \oplus K \otimes_A C_a \cong C_a^{\oplus m+1}$$

This proves (1). Finally, suppose that in addition  $A_a$  is smooth over  $R$ . Then the same presentation shows that  $\Omega_{C_a/R}$  is the cokernel of the map

$$J'/(J')^2 \longrightarrow \bigoplus_i C_a dx_i \oplus \bigoplus_j C_a dy_j$$

The summand  $C_a$  of  $J'/(J')^2$  in the decomposition above corresponds to  $hx_0 - 1$  and hence maps isomorphically to the summand  $C_a dx_0$ . The summand  $I/I^2 \otimes_A C_a$  of  $J'/(J')^2$  maps injectively to  $\bigoplus_{i=1, \dots, n} C_a dx_i$  with quotient  $\Omega_{A_a/R} \otimes_{A_a} C_a$ . The summand  $K \otimes_A C_a$  maps injectively to  $\bigoplus_{j \geq 1} C_a dy_j$  with quotient isomorphic to  $I/I^2 \otimes_A C_a$ . Thus the cokernel of the last displayed map is the module  $I/I^2 \otimes_A C_a \oplus \Omega_{A_a/R} \otimes_{A_a} C_a$ . Since  $(I/I^2)_a \oplus \Omega_{A_a/R}$  is free (from the definition of smooth ring maps) we see that (2) holds.  $\square$

The following proposition was proved for smooth ring maps over henselian pairs by Elkik in [Elk73]. For smooth ring maps it can be found in [Ara01], where it is also proven that ring maps between smooth algebras can be lifted.

**Proposition 3.2.** *Let  $R \rightarrow R_0$  be a surjective ring map with kernel  $I$ .*

- (1) *If  $R_0 \rightarrow A_0$  is a syntomic ring map, then there exists a syntomic ring map  $R \rightarrow A$  such that  $A/IA \cong A_0$ .*

- (2) If  $R_0 \rightarrow A_0$  is a smooth ring map, then there exists a smooth ring map  $R \rightarrow A$  such that  $A/IA \cong A_0$ .

**Proof.** Assume  $R_0 \rightarrow A_0$  syntomic, in particular a local complete intersection (More on Algebra, Lemma ??). Choose a presentation  $A_0 = R_0[x_1, \dots, x_n]/J_0$ . Set  $C_0 = \text{Sym}_{A_0}^*(J_0/J_0^2)$ . Note that  $J_0/J_0^2$  is a finite projective  $A_0$ -module (Algebra, Lemma ??). By Lemma 3.1 the ring map  $A_0 \rightarrow C_0$  is smooth and we can find a presentation  $C_0 = R_0[y_1, \dots, y_m]/K_0$  with  $K_0/K_0^2$  free over  $C_0$ . By Algebra, Lemma ?? we can assume  $C_0 = R_0[y_1, \dots, y_m]/(\bar{f}_1, \dots, \bar{f}_c)$  where  $\bar{f}_1, \dots, \bar{f}_c$  maps to a basis of  $K_0/K_0^2$  over  $C_0$ . Choose  $f_1, \dots, f_c \in R[y_1, \dots, y_c]$  lifting  $\bar{f}_1, \dots, \bar{f}_c$  and set

$$C = R[y_1, \dots, y_m]/(f_1, \dots, f_c)$$

By construction  $C_0 = C/IC$ . By Algebra, Lemma ?? we can after replacing  $C$  by  $C_g$  assume that  $C$  is a relative global complete intersection over  $R$ . We conclude that there exists a finite projective  $A_0$ -module  $P_0$  such that  $C_0 = \text{Sym}_{A_0}^*(P_0)$  is isomorphic to  $C/IC$  for some syntomic  $R$ -algebra  $C$ .

Choose an integer  $n$  and a direct sum decomposition  $A_0^{\oplus n} = P_0 \oplus Q_0$ . By More on Algebra, Lemma ?? we can find an étale ring map  $C \rightarrow C'$  which induces an isomorphism  $C/IC \rightarrow C'/IC'$  and a finite projective  $C'$ -module  $Q$  such that  $Q/ICQ$  is isomorphic to  $Q_0 \otimes_{A_0} C/IC$ . Then  $D = \text{Sym}_{C'}^*(Q)$  is a smooth  $C'$ -algebra (see More on Algebra, Lemma ??). Picture

$$\begin{array}{ccccccc} R & \longrightarrow & C & \longrightarrow & C' & \longrightarrow & D \\ \downarrow & & \downarrow & & \downarrow & & \downarrow \\ R/I & \longrightarrow & A_0 & \longrightarrow & C/IC & \xrightarrow{\cong} & C'/IC' \longrightarrow D/ID \end{array}$$

Observe that our choice of  $Q$  gives

$$\begin{aligned} D/ID &= \text{Sym}_{C'/IC}^*(Q_0 \otimes_{A_0} C/IC) \\ &= \text{Sym}_{A_0}^*(Q_0) \otimes_{A_0} C/IC \\ &= \text{Sym}_{A_0}^*(Q_0) \otimes_{A_0} \text{Sym}_{A_0}^*(P_0) \\ &= \text{Sym}_{A_0}^*(Q_0 \oplus P_0) \\ &= \text{Sym}_{A_0}^*(A_0^{\oplus n}) \\ &= A_0[x_1, \dots, x_n] \end{aligned}$$

Choose  $f_1, \dots, f_n \in D$  which map to  $x_1, \dots, x_n$  in  $D/ID = A_0[x_1, \dots, x_n]$ . Set  $A = D/(f_1, \dots, f_n)$ . Note that  $A_0 = A/IA$ . We claim that  $R \rightarrow A$  is syntomic in a neighbourhood of  $V(IA)$ . If the claim is true, then we can find a  $f \in A$  mapping to  $1 \in A_0$  such that  $A_f$  is syntomic over  $R$  and the proof of (1) is finished.

**Proof of the claim.** Observe that  $R \rightarrow D$  is syntomic as a composition of the syntomic ring map  $R \rightarrow C$ , the étale ring map  $C \rightarrow C'$  and the smooth ring map  $C' \rightarrow D$  (Algebra, Lemmas ?? and ??). The question is local on  $\text{Spec}(D)$ , hence we may assume that  $D$  is a relative global complete intersection (Algebra, Lemma ??). Say  $D = R[y_1, \dots, y_m]/(g_1, \dots, g_s)$ . Let  $f'_1, \dots, f'_n \in R[y_1, \dots, y_m]$  be lifts of  $f_1, \dots, f_n$ . Then we can apply Algebra, Lemma ?? to get the claim.

**Proof of (2).** Since a smooth ring map is syntomic, we can find a syntomic ring map  $R \rightarrow A$  such that  $A_0 = A/IA$ . By assumption the fibres of  $R \rightarrow A$  are smooth

over primes in  $V(I)$  hence  $R \rightarrow A$  is smooth in an open neighbourhood of  $V(IA)$  (Algebra, Lemma ??). Thus we can replace  $A$  by a localization to obtain the result we want.  $\square$

We know that any syntomic ring map  $R \rightarrow A$  is locally a relative global complete intersection, see Algebra, Lemma ??. The next lemma says that a vector bundle over  $\text{Spec}(A)$  is a relative global complete intersection.

**Lemma 3.3.** *Let  $R \rightarrow A$  be a syntomic ring map. Then there exists a smooth  $R$ -algebra map  $A \rightarrow C$  with a retraction such that  $C$  is a global relative complete intersection over  $R$ , i.e.,*

$$C \cong R[x_1, \dots, x_n]/(f_1, \dots, f_c)$$

*flat over  $R$  and all fibres of dimension  $n - c$ .*

**Proof.** Apply Lemma 3.1 to get  $A \rightarrow C$ . By Algebra, Lemma ?? we can write  $C = R[x_1, \dots, x_n]/(f_1, \dots, f_c)$  with  $f_i$  mapping to a basis of  $J/J^2$ . The ring map  $R \rightarrow C$  is syntomic (hence flat) as it is a composition of a syntomic and a smooth ring map. The dimension of the fibres is  $n - c$  by Algebra, Lemma ?? (the fibres are local complete intersections, so the lemma applies).  $\square$

**Lemma 3.4.** *Let  $R \rightarrow A$  be a smooth ring map. Then there exists a smooth  $R$ -algebra map  $A \rightarrow B$  with a retraction such that  $B$  is standard smooth over  $R$ , i.e.,*

$$B \cong R[x_1, \dots, x_n]/(f_1, \dots, f_c)$$

*and  $\det(\partial f_j / \partial x_i)_{i,j=1,\dots,c}$  is invertible in  $B$ .*

**Proof.** Apply Lemma 3.3 to get a smooth  $R$ -algebra map  $A \rightarrow C$  with a retraction such that  $C = R[x_1, \dots, x_n]/(f_1, \dots, f_c)$  is a relative global complete intersection over  $R$ . As  $C$  is smooth over  $R$  we have a short exact sequence

$$0 \rightarrow \bigoplus_{j=1,\dots,c} C f_j \rightarrow \bigoplus_{i=1,\dots,n} C dx_i \rightarrow \Omega_{C/R} \rightarrow 0$$

Since  $\Omega_{C/R}$  is a projective  $C$ -module this sequence is split. Choose a left inverse  $t$  to the first map. Say  $t(dx_i) = \sum c_{ij} f_j$  so that  $\sum_i \frac{\partial f_j}{\partial x_i} c_{i\ell} = \delta_{j\ell}$  (Kronecker delta). Let

$$B' = C[y_1, \dots, y_c] = R[x_1, \dots, x_n, y_1, \dots, y_c]/(f_1, \dots, f_c)$$

The  $R$ -algebra map  $C \rightarrow B'$  has a retraction given by mapping  $y_j$  to zero. We claim that the map

$$R[z_1, \dots, z_n] \rightarrow B', \quad z_i \mapsto x_i - \sum_j c_{ij} y_j$$

is étale at every point in the image of  $\text{Spec}(C) \rightarrow \text{Spec}(B')$ . In  $\Omega_{B'/R[z_1,\dots,z_n]}$  we have

$$0 = df_j - \sum_i \frac{\partial f_j}{\partial x_i} dz_i \equiv \sum_{i,\ell} \frac{\partial f_j}{\partial x_i} c_{i\ell} dy_\ell \equiv dy_j \pmod{(y_1, \dots, y_c) \Omega_{B'/R[z_1,\dots,z_n]}}$$

Since  $0 = dz_i = dx_i$  modulo  $\sum B' dy_j + (y_1, \dots, y_c) \Omega_{B'/R[z_1,\dots,z_n]}$  we conclude that

$$\Omega_{B'/R[z_1,\dots,z_n]}/(y_1, \dots, y_c) \Omega_{B'/R[z_1,\dots,z_n]} = 0.$$

As  $\Omega_{B'/R[z_1,\dots,z_n]}$  is a finite  $B'$ -module by Nakayama's lemma there exists a  $g \in 1 + (y_1, \dots, y_c)$  that  $(\Omega_{B'/R[z_1,\dots,z_n]})_g = 0$ . This proves that  $R[z_1, \dots, z_n] \rightarrow B'_g$  is unramified, see Algebra, Definition ??. For any ring map  $R \rightarrow k$  where  $k$  is a field

we obtain an unramified ring map  $k[z_1, \dots, z_n] \rightarrow (B'_g) \otimes_R k$  between smooth  $k$ -algebras of dimension  $n$ . It follows that  $k[z_1, \dots, z_n] \rightarrow (B'_g) \otimes_R k$  is flat by Algebra, Lemmas ?? and ?. By the critère de platitude par fibre (Algebra, Lemma ??) we conclude that  $R[z_1, \dots, z_n] \rightarrow B'_g$  is flat. Finally, Algebra, Lemma ?? implies that  $R[z_1, \dots, z_n] \rightarrow B'_g$  is étale. Set  $B = B'_g$ . Note that  $C \rightarrow B$  is smooth and has a retraction, so also  $A \rightarrow B$  is smooth and has a retraction. Moreover,  $R[z_1, \dots, z_n] \rightarrow B$  is étale. By Algebra, Lemma ?? we can write

$$B = R[z_1, \dots, z_n, w_1, \dots, w_m]/(g_1, \dots, g_m)$$

with  $\det(\partial g_j / \partial w_i)$  invertible in  $B$ . This proves the lemma.  $\square$

**Lemma 3.5.** *Let  $R \rightarrow \Lambda$  be a ring map. If  $\Lambda$  is a filtered colimit of smooth  $R$ -algebras, then  $\Lambda$  is a filtered colimit of standard smooth  $R$ -algebras.*

**Proof.** Let  $A \rightarrow \Lambda$  be an  $R$ -algebra map with  $A$  of finite presentation over  $R$ . According to Algebra, Lemma ?? we have to factor this map through a standard smooth algebra, and we know we can factor it as  $A \rightarrow B \rightarrow \Lambda$  with  $B$  smooth over  $R$ . Choose an  $R$ -algebra map  $B \rightarrow C$  with a retraction  $C \rightarrow B$  such that  $C$  is standard smooth over  $R$ , see Lemma 3.4. Then the desired factorization is  $A \rightarrow B \rightarrow C \rightarrow B \rightarrow \Lambda$ .  $\square$

**Lemma 3.6.** *Let  $R \rightarrow A$  be a standard smooth ring map. Let  $E \subset A$  be a finite subset of order  $|E| = n$ . Then there exists a presentation  $A = R[x_1, \dots, x_{n+m}]/(f_1, \dots, f_c)$  with  $c \geq n$ , with  $\det(\partial f_j / \partial x_i)_{i,j=1,\dots,c}$  invertible in  $A$ , and such that  $E$  is the set of congruence classes of  $x_1, \dots, x_n$ .*

**Proof.** Choose a presentation  $A = R[y_1, \dots, y_m]/(g_1, \dots, g_d)$  such that the image of  $\det(\partial g_j / \partial y_i)_{i,j=1,\dots,d}$  is invertible in  $A$ . Choose an enumerations  $E = \{a_1, \dots, a_n\}$  and choose  $h_i \in R[y_1, \dots, y_m]$  whose image in  $A$  is  $a_i$ . Consider the presentation

$$A = R[x_1, \dots, x_n, y_1, \dots, y_m]/(x_1 - h_1, \dots, x_n - h_n, g_1, \dots, g_d)$$

and set  $c = n + d$ .  $\square$

**Lemma 3.7.** *Let  $R \rightarrow A$  be a ring map of finite presentation. Let  $a \in A$ . Consider the following conditions on  $a$ :*

- (1)  $A_a$  is smooth over  $R$ ,
- (2)  $A_a$  is smooth over  $R$  and  $\Omega_{A_a/R}$  is stably free,
- (3)  $A_a$  is smooth over  $R$  and  $\Omega_{A_a/R}$  is free,
- (4)  $A_a$  is standard smooth over  $R$ ,
- (5)  $a$  is strictly standard in  $A$  over  $R$ ,
- (6)  $a$  is elementary standard in  $A$  over  $R$ .

Then we have

- (a) (4)  $\Rightarrow$  (3)  $\Rightarrow$  (2)  $\Rightarrow$  (1),
- (b) (6)  $\Rightarrow$  (5),
- (c) (6)  $\Rightarrow$  (4),
- (d) (5)  $\Rightarrow$  (2),
- (e) (2)  $\Rightarrow$  the elements  $a^e$ ,  $e \geq e_0$  are strictly standard in  $A$  over  $R$ ,
- (f) (4)  $\Rightarrow$  the elements  $a^e$ ,  $e \geq e_0$  are elementary standard in  $A$  over  $R$ .



**Proof.** Part (a) is clear from the definitions and Algebra, Lemma ???. Part (b) is clear from Definition 2.3.

Proof of (c). Choose a presentation  $A = R[x_1, \dots, x_n]/(f_1, \dots, f_m)$  such that (2.3.1) and (2.3.2) hold. Choose  $h \in R[x_1, \dots, x_n]$  mapping to  $a$ . Then

$$A_a = R[x_0, x_1, \dots, x_n]/(x_0h - 1, f_1, \dots, f_m).$$

Write  $J = (x_0h - 1, f_1, \dots, f_m)$ . By (2.3.2) we see that the  $A_a$ -module  $J/J^2$  is generated by  $x_0h - 1, f_1, \dots, f_c$  over  $A_a$ . Hence, as in the proof of Algebra, Lemma ???, we can choose a  $g \in 1 + J$  such that

$$A_a = R[x_0, \dots, x_n, x_{n+1}]/(x_0h - 1, f_1, \dots, f_c, gx_{n+1} - 1).$$

At this point (2.3.1) implies that  $R \rightarrow A_a$  is standard smooth (use the coordinates  $x_0, x_1, \dots, x_c, x_{n+1}$  to take derivatives).

Proof of (d). Choose a presentation  $A = R[x_1, \dots, x_n]/(f_1, \dots, f_m)$  such that (2.3.3) and (2.3.4) hold. Write  $I = (f_1, \dots, f_m)$ . We already know that  $A_a$  is smooth over  $R$ , see Lemma 2.5. By Lemma 2.4 we see that  $(I/I^2)_a$  is free on  $f_1, \dots, f_c$  and maps isomorphically to a direct summand of  $\bigoplus A_a dx_i$ . Since  $\Omega_{A_a/R} = (\Omega_{A/R})_a$  is the cokernel of the map  $(I/I^2)_a \rightarrow \bigoplus A_a dx_i$  we conclude that it is stably free.

Proof of (e). Choose a presentation  $A = R[x_1, \dots, x_n]/I$  with  $I$  finitely generated. By assumption we have a short exact sequence

$$0 \rightarrow (I/I^2)_a \rightarrow \bigoplus_{i=1, \dots, n} A_a dx_i \rightarrow \Omega_{A_a/R} \rightarrow 0$$

which is split exact. Hence we see that  $(I/I^2)_a \oplus \Omega_{A_a/R}$  is a free  $A_a$ -module. Since  $\Omega_{A_a/R}$  is stably free we see that  $(I/I^2)_a$  is stably free as well. Thus replacing the presentation chosen above by  $A = R[x_1, \dots, x_n, x_{n+1}, \dots, x_{n+r}]/J$  with  $J = (I, x_{n+1}, \dots, x_{n+r})$  for some  $r$  we get that  $(J/J^2)_a$  is (finite) free. Choose  $f_1, \dots, f_c \in J$  which map to a basis of  $(J/J^2)_a$ . Extend this to a list of generators  $f_1, \dots, f_m \in J$ . Consider the presentation  $A = R[x_1, \dots, x_{n+r}]/(f_1, \dots, f_m)$ . Then (2.3.4) holds for  $a^e$  for all sufficiently large  $e$  by construction. Moreover, since  $(J/J^2)_a \rightarrow \bigoplus_{i=1, \dots, n+r} A_a dx_i$  is a split injection we can find an  $A_a$ -linear left inverse. Writing this left inverse in terms of the basis  $f_1, \dots, f_c$  and clearing denominators we find a linear map  $\psi_0 : A^{\oplus n+r} \rightarrow A^{\oplus c}$  such that

$$A^{\oplus c} \xrightarrow{(f_1, \dots, f_c)} J/J^2 \xrightarrow{f \mapsto df} \bigoplus_{i=1, \dots, n+r} A dx_i \xrightarrow{\psi_0} A^{\oplus c}$$

is multiplication by  $a^{e_0}$  for some  $e_0 \geq 1$ . By Lemma 2.4 we see (2.3.3) holds for all  $a^{ce_0}$  and hence for  $a^e$  for all  $e$  with  $e \geq ce_0$ .

Proof of (f). Choose a presentation  $A_a = R[x_1, \dots, x_n]/(f_1, \dots, f_c)$  such that  $\det(\partial f_j / \partial x_i)_{i,j=1, \dots, c}$  is invertible in  $A_a$ . We may assume that for some  $m < n$  the classes of the elements  $x_1, \dots, x_m$  correspond  $a_i/1$  where  $a_1, \dots, a_m \in A$  are generators of  $A$  over  $R$ , see Lemma 3.6. After replacing  $x_i$  by  $a^N x_i$  for  $m < i \leq n$  we may assume the class of  $x_i$  is  $a_i/1 \in A_a$  for some  $a_i \in A$ . Consider the ring map

$$\Psi : R[x_1, \dots, x_n] \longrightarrow A, \quad x_i \longmapsto a_i.$$

This is a surjective ring map. By replacing  $f_j$  by  $a^N f_j$  we may assume that  $f_j \in R[x_1, \dots, x_n]$  and that  $\Psi(f_j) = 0$  (since after all  $f_j(a_1/1, \dots, a_n/1) = 0$  in  $A_a$ ). Let  $J = \text{Ker}(\Psi)$ . Then  $A = R[x_1, \dots, x_n]/J$  is a presentation and  $f_1, \dots, f_c \in J$

are elements such that  $(J/J^2)_a$  is freely generated by  $f_1, \dots, f_c$  and such that  $\det(\partial f_j / \partial x_i)_{i,j=1, \dots, c}$  maps to an invertible element of  $A_a$ . It follows that (2.3.1) and (2.3.2) hold for  $a^e$  and all large enough  $e$  as desired.  $\square$

#### 4. Intermezzo: Néron desingularization

We interrupt the attack on the general case of Popescu's theorem to an easier but already very interesting case, namely, when  $R \rightarrow \Lambda$  is a homomorphism of discrete valuation rings. This is discussed in [Art69, Section 4].

**Situation 4.1.** Here  $R \subset \Lambda$  is an extension of discrete valuation rings with ramification index 1 (More on Algebra, Definition ??). We assume given a factorization

$$R \rightarrow A \xrightarrow{\varphi} \Lambda$$

with  $R \rightarrow A$  flat and of finite type. Let  $\mathfrak{q} = \text{Ker}(\varphi)$  and  $\mathfrak{p} = \varphi^{-1}(\mathfrak{m}_\Lambda)$ .

In Situation 4.1 let  $\pi \in R$  be a uniformizer. Recall that flatness of  $A$  over  $R$  signifies that  $\pi$  is a nonzerodivisor on  $A$  (More on Algebra, Lemma ??). By our assumption on  $R \subset \Lambda$  we see that  $\pi$  maps to a uniformizer of  $\Lambda$ . Since  $\pi \in \mathfrak{p}$  we can consider Néron's affine blowup algebra (see Algebra, Section ??)

$$\varphi' : A' = A[\frac{\mathfrak{p}}{\pi}] \longrightarrow \Lambda$$

which comes endowed with an induced map to  $\Lambda$  sending  $a/\pi^n$ ,  $a \in \mathfrak{p}^n$  to  $\pi^{-n}\varphi(a)$  in  $\Lambda$ . We will denote  $\mathfrak{q}' \subset \mathfrak{p}' \subset A'$  the corresponding prime ideals of  $A'$ . Observe that the isomorphism class of  $A'$  does not depend on our choice of uniformizer. Repeating the construction we obtain a sequence

$$A \rightarrow A' \rightarrow A'' \rightarrow \dots \rightarrow \Lambda$$

**Lemma 4.2.** *In Situation 4.1 Néron's blowup is functorial in the following sense*

- (1) *if  $a \in A$ ,  $a \notin \mathfrak{p}$ , then Néron's blowup of  $A_a$  is  $A'_a$ , and*
- (2) *if  $B \rightarrow A$  is a surjection of flat finite type  $R$ -algebras with kernel  $I$ , then  $A'$  is the quotient of  $B'/IB'$  by its  $\pi$ -power torsion.*

**Proof.** Both (1) and (2) are special cases of Algebra, Lemma ???. In fact, whenever we have  $A_1 \rightarrow A_2 \rightarrow \Lambda$  such that  $\mathfrak{p}_1 A_2 = \mathfrak{p}_2$ , we have that  $A'_2$  is the quotient of  $A'_1 \otimes_{A_1} A_2$  by its  $\pi$ -power torsion.  $\square$

**Lemma 4.3.** *In Situation 4.1 assume that  $R \rightarrow A$  is smooth at  $\mathfrak{p}$  and that  $R/\pi R \subset \Lambda/\pi\Lambda$  is a separable field extension. Then  $R \rightarrow A'$  is smooth at  $\mathfrak{p}'$  and there is a short exact sequence*

$$0 \rightarrow \Omega_{A/R} \otimes_A A'_{\mathfrak{p}'} \rightarrow \Omega_{A'/R, \mathfrak{p}'} \rightarrow (A'/\pi A')_{\mathfrak{p}'}^{\oplus c} \rightarrow 0$$

where  $c = \dim((A/\pi A)_{\mathfrak{p}})$ .

**Proof.** By Lemma 4.2 we may replace  $A$  by a localization at an element not in  $\mathfrak{p}$ ; we will use this without further mention. Write  $\kappa = R/\pi R$ . Since smoothness is stable under base change (Algebra, Lemma ??) we see that  $A/\pi A$  is smooth over  $\kappa$  at  $\mathfrak{p}$ . Hence  $(A/\pi A)_{\mathfrak{p}}$  is a regular local ring (Algebra, Lemma ??). Choose  $g_1, \dots, g_c \in \mathfrak{p}$  which map to a regular system of parameters in  $(A/\pi A)_{\mathfrak{p}}$ . Then we see that  $\mathfrak{p} = (\pi, g_1, \dots, g_c)$  after possibly replacing  $A$  by a localization. Note that  $\pi, g_1, \dots, g_c$  is a regular sequence in  $A_{\mathfrak{p}}$  (first  $\pi$  is a nonzerodivisor and then Algebra, Lemma ?? for the rest of the sequence). After replacing  $A$  by a localization

we may assume that  $\pi, g_1, \dots, g_c$  is a regular sequence in  $A$  (Algebra, Lemma ??). It follows that

$$A' = A[y_1, \dots, y_c]/(\pi y_1 - g_1, \dots, \pi y_c - g_c) = A[y_1, \dots, y_c]/I$$

by More on Algebra, Lemma ?. In the following we will use the definition of smoothness using the naive cotangent complex (Algebra, Definition ??) and the criterion of Algebra, Lemma ?? without further mention. The exact sequence of Algebra, Lemma ?? for  $R \rightarrow A[y_1, \dots, y_c] \rightarrow A'$  looks like this

$$0 \rightarrow H_1(NL_{A'/R}) \rightarrow I/I^2 \rightarrow \Omega_{A/R} \otimes_A A' \oplus \bigoplus_{i=1, \dots, c} A' dy_i \rightarrow \Omega_{A'/R} \rightarrow 0$$

where the class of  $\pi y_i - g_i$  in  $I/I^2$  is mapped to  $-dg_i + \pi dy_i$  in the next term. Here we have used Algebra, Lemma ?? to compute  $NL_{A'/A[y_1, \dots, y_c]}$  and we have used that  $R \rightarrow A[y_1, \dots, y_c]$  is smooth, so  $H_1(NL_{A[y_1, \dots, y_c]/R}) = 0$  and  $\Omega_{A[y_1, \dots, y_c]/R}$  is a finite projective (a fortiori flat)  $A[y_1, \dots, y_c]$ -module which is in fact the direct sum of  $\Omega_{A/R} \otimes_A A[y_1, \dots, y_c]$  and a free module with basis  $dy_i$ . To finish the proof it suffices to show that  $dg_1, \dots, dg_c$  forms part of a basis for the finite free module  $\Omega_{A/R, \mathfrak{p}}$ . Namely, this will show  $(I/I^2)_{\mathfrak{p}}$  is free on  $\pi y_i - g_i$ , the localization at  $\mathfrak{p}$  of the middle map in the sequence is injective, so  $H_1(NL_{A'/R})_{\mathfrak{p}} = 0$ , and that the cokernel  $\Omega_{A'/R, \mathfrak{p}}$  is finite free. To do this it suffices to show that the images of  $dg_i$  are  $\kappa(\mathfrak{p})$ -linearly independent in  $\Omega_{A/R, \mathfrak{p}}/\pi = \Omega_{(A/\pi A)/\kappa, \mathfrak{p}}$  (equality by Algebra, Lemma ??). Since  $\kappa \subset \kappa(\mathfrak{p}) \subset \Lambda/\pi\Lambda$  we see that  $\kappa(\mathfrak{p})$  is separable over  $\kappa$  (Algebra, Definition ??). The desired linear independence now follows from Algebra, Lemma ??.  $\square$

**Lemma 4.4.** *In Situation 4.1 assume that  $R \rightarrow A$  is smooth at  $\mathfrak{q}$  and that we have a surjection of  $R$ -algebras  $B \rightarrow A$  with kernel  $I$ . Assume  $R \rightarrow B$  smooth at  $\mathfrak{p}_B = (B \rightarrow A)^{-1}\mathfrak{p}$ . If the cokernel of*

$$I/I^2 \otimes_A \Lambda \rightarrow \Omega_{B/R} \otimes_B \Lambda$$

*is a free  $\Lambda$ -module, then  $R \rightarrow A$  is smooth at  $\mathfrak{p}$ .*

**Proof.** The cokernel of the map  $I/I^2 \rightarrow \Omega_{B/R} \otimes_B A$  is  $\Omega_{A/R}$ , see Algebra, Lemma ??. Let  $d = \dim_{\mathfrak{q}}(A/R)$  be the relative dimension of  $R \rightarrow A$  at  $\mathfrak{q}$ , i.e., the dimension of  $\text{Spec}(A[1/\pi])$  at  $\mathfrak{q}$ . See Algebra, Definition ??. Then  $\Omega_{A/R, \mathfrak{q}}$  is free over  $A_{\mathfrak{q}}$  of rank  $d$  (Algebra, Lemma ??). Thus if the hypothesis of the lemma holds, then  $\Omega_{A/R} \otimes_A \Lambda$  is free of rank  $d$ . It follows that  $\Omega_{A/R} \otimes_A \kappa(\mathfrak{p})$  has dimension  $d$  (as it is true upon tensoring with  $\Lambda/\pi\Lambda$ ). Since  $R \rightarrow A$  is flat and since  $\mathfrak{p}$  is a specialization of  $\mathfrak{q}$ , we see that  $\dim_{\mathfrak{p}}(A/R) \geq d$  by Algebra, Lemma ??. Then it follows that  $R \rightarrow A$  is smooth at  $\mathfrak{p}$  by Algebra, Lemmas ?? and ??.  $\square$

**Lemma 4.5.** *In Situation 4.1 assume that  $R \rightarrow A$  is smooth at  $\mathfrak{q}$  and that  $R/\pi R \subset \Lambda/\pi\Lambda$  is a separable extension of fields. Then after a finite number of affine Néron blowups the algebra  $A$  becomes smooth over  $R$  at  $\mathfrak{p}$ .*

**Proof.** We choose an  $R$ -algebra  $B$  and a surjection  $B \rightarrow A$ . Set  $\mathfrak{p}_B = (B \rightarrow A)^{-1}(\mathfrak{p})$  and denote  $r$  the relative dimension of  $R \rightarrow B$  at  $\mathfrak{p}_B$ . We choose  $B$  such that  $R \rightarrow B$  is smooth at  $\mathfrak{p}_B$ . For example we can take  $B$  to be a polynomial algebra in  $r$  variables over  $R$ . Consider the complex

$$I/I^2 \otimes_A \Lambda \longrightarrow \Omega_{B/R} \otimes_B \Lambda$$

of Lemma 4.4. By the structure of finite modules over  $\Lambda$  (More on Algebra, Lemma ??) we see that the cokernel looks like

$$\Lambda^{\oplus d} \oplus \bigoplus_{i=1, \dots, n} \Lambda / \pi^{e_i} \Lambda$$

for some  $d \geq 0$ ,  $n \geq 0$ , and  $e_i \geq 1$ . Observe that  $d$  is the relative dimension of  $A/R$  at  $\mathfrak{q}$  (Algebra, Lemma ??). If the defect  $e = \sum_{i=1, \dots, n} e_i$  is zero, then we are done by Lemma 4.4.

Next, we consider what happens when we perform the Néron blowup. Recall that  $A'$  is the quotient of  $B'/IB'$  by its  $\pi$ -power torsion (Lemma 4.2) and that  $R \rightarrow B'$  is smooth at  $\mathfrak{p}_{B'}$  (Lemma 4.3). Thus after blowup we have exactly the same setup. Picture

$$\begin{array}{ccccccc} 0 & \longrightarrow & I' & \longrightarrow & B' & \longrightarrow & A' \longrightarrow 0 \\ & & \uparrow & & \uparrow & & \uparrow \\ 0 & \longrightarrow & I & \longrightarrow & B & \longrightarrow & A \longrightarrow 0 \end{array}$$

Since  $I \subset \mathfrak{p}_B$ , we see that  $I \rightarrow I'$  factors through  $\pi I'$ . Looking at the induced map of complexes we get

$$\begin{array}{ccc} I'/(I')^2 \otimes_{A'} \Lambda & \longrightarrow & \Omega_{B'/R} \otimes_{B'} \Lambda = M' \\ \uparrow & & \uparrow \\ I/I^2 \otimes_A \Lambda & \longrightarrow & \Omega_{B/R} \otimes_B \Lambda = M \end{array}$$

Then  $M \subset M'$  are finite free  $\Lambda$ -modules with quotient  $M'/M$  annihilated by  $\pi$ , see Lemma 4.3. Let  $N \subset M$  and  $N' \subset M'$  be the images of the horizontal maps and denote  $Q = M/N$  and  $Q' = M'/N'$ . We obtain a commutative diagram

$$\begin{array}{ccccccc} 0 & \longrightarrow & N' & \longrightarrow & M' & \longrightarrow & Q' \longrightarrow 0 \\ & & \uparrow & & \uparrow & & \uparrow \\ 0 & \longrightarrow & N & \longrightarrow & M & \longrightarrow & Q \longrightarrow 0 \end{array}$$

Then  $N \subset N'$  are free  $\Lambda$ -modules of rank  $r - d$ . Since  $I$  maps into  $\pi I'$  we see that  $N \subset \pi N'$ .

Let  $K = \Lambda_\pi$  be the fraction field of  $\Lambda$ . We have a commutative diagram

$$\begin{array}{ccccccc} 0 & \longrightarrow & N' & \longrightarrow & N'_K \cap M' & \longrightarrow & Q'_{tor} \longrightarrow 0 \\ & & \uparrow & & \uparrow & & \uparrow \\ 0 & \longrightarrow & N & \longrightarrow & N_K \cap M & \longrightarrow & Q_{tor} \longrightarrow 0 \end{array}$$

whose rows are short exact sequences. This shows that the change in defect is given by

$$e - e' = \text{length}(Q_{tor}) - \text{length}(Q'_{tor}) = \text{length}(N'/N) - \text{length}(N'_K \cap M' / N_K \cap M)$$

Since  $M'/M$  is annihilated by  $\pi$ , so is  $N'_K \cap M' / N_K \cap M$ , and its length is at most  $\dim_K(N_K)$ . Since  $N \subset \pi N'$  we get  $\text{length}(N'/N) \geq \dim_K(N_K)$ , with equality if and only if  $N = \pi N'$ .

To finish the proof we have to show that  $N$  is strictly smaller than  $\pi N'$  when  $A$  is not smooth at  $\mathfrak{p}$ ; this is the key computation one has to do in Néron's argument. To do this, we consider the exact sequence

$$I/I^2 \otimes_B \kappa(\mathfrak{p}_B) \rightarrow \Omega_{B/R} \otimes_B \kappa(\mathfrak{p}_B) \rightarrow \Omega_{A/R} \otimes_A \kappa(\mathfrak{p}) \rightarrow 0$$

(follows from Algebra, Lemma ??). Since  $R \rightarrow A$  is not smooth at  $\mathfrak{p}$  we see that the dimension  $s$  of  $\Omega_{A/R} \otimes_A \kappa(\mathfrak{p})$  is bigger than  $d$ . On the other hand the first arrow factors through the injective map

$$\mathfrak{p}B_{\mathfrak{p}}/\mathfrak{p}^2B_{\mathfrak{p}} \rightarrow \Omega_{B/R} \otimes_B \kappa(\mathfrak{p}_B)$$

of Algebra, Lemma ??; note that  $\kappa(\mathfrak{p})$  is separable over  $k$  by our assumption on  $R/\pi R \subset \Lambda/\pi\Lambda$ . Hence we conclude that we can find generators  $g_1, \dots, g_t \in I$  such that  $g_j \in \mathfrak{p}^2$  for  $j > r - s$ . Then the images of  $g_j$  in  $A'$  are in  $\pi^2 I'$  for  $j > r - s$ . Since  $r - s < r - d$  we find that at least one of the minimal generators of  $N$  becomes divisible by  $\pi^2$  in  $N'$ . Thus we see that  $e$  decreases by at least 1 and we win.  $\square$

If  $R \rightarrow \Lambda$  is an extension of discrete valuation rings, then  $R \rightarrow \Lambda$  is regular if and only if (a) the ramification index is 1, (b) the extension of fraction fields is separable, and (c)  $R/\mathfrak{m}_R \subset \Lambda/\mathfrak{m}_\Lambda$  is separable. Thus the following result is a special case of general Néron desingularization in Theorem 12.1.

**Lemma 4.6.** *Let  $R \subset \Lambda$  be an extension of discrete valuation rings which has ramification index 1 and induces a separable extension of residue fields and of fraction fields. Then  $\Lambda$  is a filtered colimit of smooth  $R$ -algebras.*

**Proof.** By Algebra, Lemma ?? it suffices to show that any  $R \rightarrow A \rightarrow \Lambda$  as in Situation 4.1 can be factored as  $A \rightarrow B \rightarrow \Lambda$  with  $B$  a smooth  $R$ -algebra. After replacing  $A$  by its image in  $\Lambda$  we may assume that  $A$  is a domain whose fraction field  $K$  is a subfield of the fraction field of  $\Lambda$ . In particular,  $A$  is separable over the fraction field of  $R$  by our assumptions. Then  $R \rightarrow A$  is smooth at  $\mathfrak{q} = (0)$  by Algebra, Lemma ??. After a finite number of Néron blowups, we may assume  $R \rightarrow A$  is smooth at  $\mathfrak{p}$ , see Lemma 4.5. Then, after replacing  $A$  by a localization at an element  $a \in A$ ,  $a \notin \mathfrak{p}$  it becomes smooth over  $R$  and the lemma is proved.  $\square$

## 5. The lifting problem

The goal in this section is to prove (Proposition 5.3) that the collection of algebras which are filtered colimits of smooth algebras is closed under infinitesimal flat deformations. The proof is elementary and only uses the results on presentations of smooth algebras from Section 3.

**Lemma 5.1.** *Let  $R \rightarrow \Lambda$  be a ring map. Let  $I \subset R$  be an ideal. Assume that*

- (1)  $I^2 = 0$ , and
- (2)  $\Lambda/I\Lambda$  is a filtered colimit of smooth  $R/I$ -algebras.

*Let  $\varphi : A \rightarrow \Lambda$  be an  $R$ -algebra map with  $A$  of finite presentation over  $R$ . Then there exists a factorization*

$$A \rightarrow B/J \rightarrow \Lambda$$

*where  $B$  is a smooth  $R$ -algebra and  $J \subset IB$  is a finitely generated ideal.*

**Proof.** Choose a factorization

$$A/IA \rightarrow \bar{B} \rightarrow \Lambda/IA$$

with  $\bar{B}$  standard smooth over  $R/I$ ; this is possible by assumption and Lemma 3.5. Write

$$\bar{B} = A/IA[t_1, \dots, t_r]/(\bar{g}_1, \dots, \bar{g}_s)$$

and say  $\bar{B} \rightarrow \Lambda/IA$  maps  $t_i$  to the class of  $\lambda_i$  modulo  $IA$ . Choose  $g_1, \dots, g_s \in A[t_1, \dots, t_r]$  lifting  $\bar{g}_1, \dots, \bar{g}_s$ . Write  $\varphi(g_i)(\lambda_1, \dots, \lambda_r) = \sum \epsilon_{ij} \mu_{ij}$  for some  $\epsilon_{ij} \in I$  and  $\mu_{ij} \in \Lambda$ . Define

$$A' = A[t_1, \dots, t_r, \delta_{i,j}]/(g_i - \sum \epsilon_{ij} \delta_{ij})$$

and consider the map

$$A' \longrightarrow \Lambda, \quad a \longmapsto \varphi(a), \quad t_i \longmapsto \lambda_i, \quad \delta_{ij} \longmapsto \mu_{ij}$$

We have

$$A'/IA' = A/IA[t_1, \dots, t_r]/(\bar{g}_1, \dots, \bar{g}_s)[\delta_{ij}] \cong \bar{B}[\delta_{ij}]$$

This is a standard smooth algebra over  $R/I$  as  $\bar{B}$  is standard smooth. Choose a presentation  $A'/IA' = R/I[x_1, \dots, x_n]/(\bar{f}_1, \dots, \bar{f}_c)$  with  $\det(\partial \bar{f}_j / \partial x_i)_{i,j=1, \dots, c}$  invertible in  $A'/IA'$ . Choose lifts  $f_1, \dots, f_c \in R[x_1, \dots, x_n]$  of  $\bar{f}_1, \dots, \bar{f}_c$ . Then

$$B = R[x_1, \dots, x_n, x_{n+1}]/(f_1, \dots, f_c, x_{n+1} \det(\partial f_j / \partial x_i)_{i,j=1, \dots, c} - 1)$$

is smooth over  $R$ . Since smooth ring maps are formally smooth (Algebra, Proposition ??) there exists an  $R$ -algebra map  $B \rightarrow A'$  which is an isomorphism modulo  $I$ . Then  $B \rightarrow A'$  is surjective by Nakayama's lemma (Algebra, Lemma ??). Thus  $A' = B/J$  with  $J \subset IB$  finitely generated (see Algebra, Lemma ??).  $\square$

**Lemma 5.2.** *Let  $R \rightarrow \Lambda$  be a ring map. Let  $I \subset R$  be an ideal. Assume that*

- (1)  $I^2 = 0$ ,
- (2)  $\Lambda/IA$  is a filtered colimit of smooth  $R/I$ -algebras, and
- (3)  $R \rightarrow \Lambda$  is flat.

*Let  $\varphi : B \rightarrow \Lambda$  be an  $R$ -algebra map with  $B$  smooth over  $R$ . Let  $J \subset IB$  be a finitely generated ideal such that  $\varphi(J) = 0$ . Then there exists  $R$ -algebra maps*

$$B \xrightarrow{\alpha} B' \xrightarrow{\beta} \Lambda$$

*such that  $B'$  is smooth over  $R$ , such that  $\alpha(J) = 0$  and such that  $\beta \circ \alpha = \varphi$ .*

**Proof.** If we can prove the lemma in case  $J = (h)$ , then we can prove the lemma by induction on the number of generators of  $J$ . Namely, suppose that  $J$  can be generated by  $n$  elements  $h_1, \dots, h_n$  and the lemma holds for all cases where  $J$  is generated by  $n-1$  elements. Then we apply the case  $n=1$  to produce  $B \rightarrow B' \rightarrow \Lambda$  where the first map kills  $h_n$ . Then we let  $J'$  be the ideal of  $B'$  generated by the images of  $h_1, \dots, h_{n-1}$  and we apply the case for  $n-1$  to produce  $B' \rightarrow B'' \rightarrow \Lambda$ . It is easy to verify that  $B \rightarrow B'' \rightarrow \Lambda$  does the job.

Assume  $J = (h)$  and write  $h = \sum \epsilon_i b_i$  for some  $\epsilon_i \in I$  and  $b_i \in B$ . Note that  $0 = \varphi(h) = \sum \epsilon_i \varphi(b_i)$ . As  $\Lambda$  is flat over  $R$ , the equational criterion for flatness (Algebra, Lemma ??) implies that we can find  $\lambda_j \in \Lambda$ ,  $j = 1, \dots, m$  and  $a_{ij} \in R$  such that  $\varphi(b_i) = \sum_j a_{ij} \lambda_j$  and  $\sum_i \epsilon_i a_{ij} = 0$ . Set

$$C = B[x_1, \dots, x_m]/(b_i - \sum a_{ij} x_j)$$

with  $C \rightarrow \Lambda$  given by  $\varphi$  and  $x_j \mapsto \lambda_j$ . Choose a factorization

$$C \rightarrow B'/J' \rightarrow \Lambda$$

as in Lemma 5.1. Since  $B$  is smooth over  $R$  we can lift the map  $B \rightarrow C \rightarrow B'/J'$  to a map  $\alpha : B \rightarrow B'$ . Then  $\varphi = \beta \circ \alpha$ . To finish the proof we check that  $\alpha(h) = 0$ . Namely, the fact that  $\alpha$  lifts  $B \rightarrow C \rightarrow B'/J'$  implies that

$$\alpha(b_i) = \sum a_{ij}\xi_j + \theta_i$$

for some  $\xi_j \in B'$  and  $\theta_i \in J' \subset IB'$ . Hence we see that

$$\alpha(h) = \alpha\left(\sum \epsilon_i b_i\right) = \sum \epsilon_i a_{ij}\xi_j + \sum \epsilon_i \theta_i = 0$$

because of the relations above and the fact that  $I^2 = 0$ .  $\square$

**Proposition 5.3.** *Let  $R \rightarrow \Lambda$  be a ring map. Let  $I \subset R$  be an ideal. Assume that*

- (1)  *$I$  is nilpotent,*
- (2)  *$\Lambda/I\Lambda$  is a filtered colimit of smooth  $R/I$ -algebras, and*
- (3)  *$R \rightarrow \Lambda$  is flat.*

*Then  $\Lambda$  is a filtered colimit of smooth  $R$ -algebras.*

**Proof.** Since  $I^n = 0$  for some  $n$ , it follows by induction on  $n$  that it suffices to consider the case where  $I^2 = 0$ . Let  $\varphi : A \rightarrow \Lambda$  be an  $R$ -algebra map with  $A$  of finite presentation over  $R$ . We have to find a factorization  $A \rightarrow B \rightarrow \Lambda$  with  $B$  smooth over  $R$ , see Algebra, Lemma ?? . By Lemma 5.1 we may assume that  $A = B/J$  with  $B$  smooth over  $R$  and  $J \subset IB$  a finitely generated ideal. By Lemma 5.2 we can find a commutative diagram

$$\begin{array}{ccc} B & \xrightarrow{\quad \alpha \quad} & B' \\ & \searrow \varphi & \swarrow \beta \\ & \Lambda & \end{array}$$

of  $R$ -algebras with  $B'$  smooth over  $R$  such that  $\alpha(J) = 0$ . Thus  $\alpha$  factors as  $B \rightarrow A \rightarrow B'$  and the proof is complete.  $\square$

## 6. The lifting lemma

Here is a fiendishly clever lemma.

**Lemma 6.1.** *Let  $R$  be a Noetherian ring. Let  $\Lambda$  be an  $R$ -algebra. Let  $\pi \in R$  and assume that  $\text{Ann}_R(\pi) = \text{Ann}_R(\pi^2)$  and  $\text{Ann}_\Lambda(\pi) = \text{Ann}_\Lambda(\pi^2)$ . Suppose we have  $R$ -algebra maps  $R/\pi^2 R \rightarrow \bar{C} \rightarrow \Lambda/\pi^2 \Lambda$  with  $\bar{C}$  of finite presentation. Then there exists an  $R$ -algebra homomorphism  $D \rightarrow \Lambda$  and a commutative diagram*

$$\begin{array}{ccccc} R/\pi^2 R & \longrightarrow & \bar{C} & \longrightarrow & \Lambda/\pi^2 \Lambda \\ \downarrow & & \downarrow & & \downarrow \\ R/\pi R & \longrightarrow & D/\pi D & \longrightarrow & \Lambda/\pi \Lambda \end{array}$$

*with the following properties*

- (a)  *$D$  is of finite presentation,*
- (b)  *$R \rightarrow D$  is smooth at any prime  $\mathfrak{q}$  with  $\pi \notin \mathfrak{q}$ ,*
- (c)  *$R \rightarrow D$  is smooth at any prime  $\mathfrak{q}$  with  $\pi \in \mathfrak{q}$  lying over a prime of  $\bar{C}$  where  $R/\pi^2 R \rightarrow \bar{C}$  is smooth, and*
- (d)  *$\bar{C}/\pi \bar{C} \rightarrow D/\pi D$  is smooth at any prime lying over a prime of  $\bar{C}$  where  $R/\pi^2 R \rightarrow \bar{C}$  is smooth.*

**Proof.** We choose a presentation

$$\bar{C} = R[x_1, \dots, x_n]/(f_1, \dots, f_m)$$

We also denote  $I = (f_1, \dots, f_m)$  and  $\bar{I}$  the image of  $I$  in  $R/\pi^2 R[x_1, \dots, x_n]$ . Since  $R$  is Noetherian, so is  $\bar{C}$ . Hence the smooth locus of  $R/\pi^2 R \rightarrow \bar{C}$  is quasi-compact, see Topology, Lemma ???. Applying Lemma 2.2 we may choose a finite list of elements  $a_1, \dots, a_r \in R[x_1, \dots, x_n]$  such that

- (1) the union of the open subspaces  $\text{Spec}(\bar{C}_{a_k}) \subset \text{Spec}(\bar{C})$  cover the smooth locus of  $R/\pi^2 R \rightarrow \bar{C}$ , and
- (2) for each  $k = 1, \dots, r$  there exists a finite subset  $E_k \subset \{1, \dots, m\}$  such that  $(\bar{I}/\bar{I}^2)_{a_k}$  is freely generated by the classes of  $f_j$ ,  $j \in E_k$ .

Set  $I_k = (f_j, j \in E_k) \subset I$  and denote  $\bar{I}_k$  the image of  $I_k$  in  $R/\pi^2 R[x_1, \dots, x_n]$ . By (2) and Nakayama's lemma we see that  $(\bar{I}/\bar{I}_k)_{a_k}$  is annihilated by  $1 + b'_k$  for some  $b'_k \in \bar{I}_{a_k}$ . Suppose  $b'_k$  is the image of  $b_k/(a_k)^N$  for some  $b_k \in I$  and some integer  $N$ . After replacing  $a_k$  by  $a_k b_k$  we get

$$(3) \quad (\bar{I}_k)_{a_k} = (\bar{I})_{a_k}.$$

Thus, after possibly replacing  $a_k$  by a high power, we may write

$$(4) \quad a_k f_\ell = \sum_{j \in E_k} h_{k,\ell}^j f_j + \pi^2 g_{k,\ell}$$

for any  $\ell \in \{1, \dots, m\}$  and some  $h_{i,\ell}^j, g_{i,\ell} \in R[x_1, \dots, x_n]$ . If  $\ell \in E_k$  we choose  $h_{k,\ell}^j = a_k \delta_{\ell,j}$  (Kronecker delta) and  $g_{k,\ell} = 0$ . Set

$$D = R[x_1, \dots, x_n, z_1, \dots, z_m]/(f_j - \pi z_j, p_{k,\ell}).$$

Here  $j \in \{1, \dots, m\}$ ,  $k \in \{1, \dots, r\}$ ,  $\ell \in \{1, \dots, m\}$ , and

$$p_{k,\ell} = a_k z_\ell - \sum_{j \in E_k} h_{k,\ell}^j z_j - \pi g_{k,\ell}.$$

Note that for  $\ell \in E_k$  we have  $p_{k,\ell} = 0$  by our choices above.

The map  $R \rightarrow D$  is the given one. Say  $\bar{C} \rightarrow \Lambda/\pi^2 \Lambda$  maps  $x_i$  to the class of  $\lambda_i$  modulo  $\pi^2$ . For an element  $f \in R[x_1, \dots, x_n]$  we denote  $f(\lambda) \in \Lambda$  the result of substituting  $\lambda_i$  for  $x_i$ . Then we know that  $f_j(\lambda) = \pi^2 \mu_j$  for some  $\mu_j \in \Lambda$ . Define  $D \rightarrow \Lambda$  by the rules  $x_i \mapsto \lambda_i$  and  $z_j \mapsto \pi \mu_j$ . This is well defined because

$$\begin{aligned} p_{k,\ell} &\mapsto a_k(\lambda) \pi \mu_\ell - \sum_{j \in E_k} h_{k,\ell}^j(\lambda) \pi \mu_j - \pi g_{k,\ell}(\lambda) \\ &= \pi \left( a_k(\lambda) \mu_\ell - \sum_{j \in E_k} h_{k,\ell}^j(\lambda) \mu_j - g_{k,\ell}(\lambda) \right) \end{aligned}$$

Substituting  $x_i = \lambda_i$  in (4) above we see that the expression inside the brackets is annihilated by  $\pi^2$ , hence it is annihilated by  $\pi$  as we have assumed  $\text{Ann}_\Lambda(\pi) = \text{Ann}_\Lambda(\pi^2)$ . The map  $\bar{C} \rightarrow D/\pi D$  is determined by  $x_i \mapsto \lambda_i$  (clearly well defined). Thus we are done if we can prove (b), (c), and (d).

Using (4) we obtain the following key equality

$$\begin{aligned} \pi p_{k,\ell} &= \pi a_k z_\ell - \sum_{j \in E_k} \pi h_{k,\ell}^j z_j - \pi^2 g_{k,\ell} \\ &= -a_k(f_\ell - \pi z_\ell) + a_k f_\ell + \sum_{j \in E_k} h_{k,\ell}^j(f_j - \pi z_j) - \sum_{j \in E_k} h_{k,\ell}^j f_j - \pi^2 g_{k,\ell} \\ &= -a_k(f_\ell - \pi z_\ell) + \sum_{j \in E_k} h_{k,\ell}^j(f_j - \pi z_j) \end{aligned}$$



The end result is an element of the ideal generated by  $f_j - \pi z_j$ . In particular, we see that  $D[1/\pi]$  is isomorphic to  $R[1/\pi][x_1, \dots, x_n, z_1, \dots, z_m]/(f_j - \pi z_j)$  which is isomorphic to  $R[1/\pi][x_1, \dots, x_n]$  hence smooth over  $R$ . This proves (b).

For fixed  $k \in \{1, \dots, r\}$  consider the ring

$$D_k = R[x_1, \dots, x_n, z_1, \dots, z_m]/(f_j - \pi z_j, j \in E_k, p_{k,\ell})$$

The number of equations is  $m = |E_k| + (m - |E_k|)$  as  $p_{k,\ell}$  is zero if  $\ell \in E_k$ . Also, note that

$$\begin{aligned} (D_k/\pi D_k)_{a_k} &= R/\pi R[x_1, \dots, x_n, 1/a_k, z_1, \dots, z_m]/(f_j, j \in E_k, p_{k,\ell}) \\ &= (\bar{C}/\pi \bar{C})_{a_k}[z_1, \dots, z_m]/(a_k z_\ell - \sum_{j \in E_k} h_{k,\ell}^j z_j) \\ &\cong (\bar{C}/\pi \bar{C})_{a_k}[z_j, j \in E_k] \end{aligned}$$

In particular  $(D_k/\pi D_k)_{a_k}$  is smooth over  $(\bar{C}/\pi \bar{C})_{a_k}$ . By our choice of  $a_k$  we have that  $(\bar{C}/\pi \bar{C})_{a_k}$  is smooth over  $R/\pi R$  of relative dimension  $n - |E_k|$ , see (2). Hence for a prime  $\mathfrak{q}_k \subset D_k$  containing  $\pi$  and lying over  $\text{Spec}(\bar{C}_{a_k})$  the fibre ring of  $R \rightarrow D_k$  is smooth at  $\mathfrak{q}_k$  of dimension  $n$ . Thus  $R \rightarrow D_k$  is syntomic at  $\mathfrak{q}_k$  by our count of the number of equations above, see Algebra, Lemma ???. Hence  $R \rightarrow D_k$  is smooth at  $\mathfrak{q}_k$ , see Algebra, Lemma ???.

To finish the proof, let  $\mathfrak{q} \subset D$  be a prime containing  $\pi$  lying over a prime where  $R/\pi^2 R \rightarrow \bar{C}$  is smooth. Then  $a_k \notin \mathfrak{q}$  for some  $k$  by (1). We will show that the surjection  $D_k \rightarrow D$  induces an isomorphism on local rings at  $\mathfrak{q}$ . Since we know that the ring maps  $\bar{C}/\pi \bar{C} \rightarrow D_k/\pi D_k$  and  $R \rightarrow D_k$  are smooth at the corresponding prime  $\mathfrak{q}_k$  by the preceding paragraph this will prove (c) and (d) and thus finish the proof.

First, note that for any  $\ell$  the equation  $\pi p_{k,\ell} = -a_k(f_\ell - \pi z_\ell) + \sum_{j \in E_k} h_{k,\ell}^j(f_j - \pi z_j)$  proved above shows that  $f_\ell - \pi z_\ell$  maps to zero in  $(D_k)_{a_k}$  and in particular in  $(D_k)_{\mathfrak{q}_k}$ . The relations (4) imply that  $a_k f_\ell = \sum_{j \in E_k} h_{k,\ell}^j f_j$  in  $I/I^2$ . Since  $(\bar{I}_k/\bar{I}_k^2)_{a_k}$  is free on  $f_j, j \in E_k$  we see that

$$a_{k'} h_{k,\ell}^j - \sum_{j' \in E_{k'}} h_{k',\ell}^{j'} h_{k,j'}^j$$

is zero in  $\bar{C}_{a_k}$  for every  $k, k', \ell$  and  $j \in E_k$ . Hence we can find a large integer  $N$  such that

$$a_k^N \left( a_{k'} h_{k,\ell}^j - \sum_{j' \in E_{k'}} h_{k',\ell}^{j'} h_{k,j'}^j \right)$$

is in  $I_k + \pi^2 R[x_1, \dots, x_n]$ . Computing modulo  $\pi$  we have

$$\begin{aligned} &a_k p_{k',\ell} - a_{k'} p_{k,\ell} + \sum h_{k',\ell}^{j'} p_{k,j'} \\ &= -a_k \sum h_{k',\ell}^{j'} z_{j'} + a_{k'} \sum h_{k,\ell}^j z_j + \sum h_{k',\ell}^{j'} a_k z_{j'} - \sum \sum h_{k',\ell}^{j'} h_{k,j'}^j z_j \\ &= \sum \left( a_{k'} h_{k,\ell}^j - \sum h_{k',\ell}^{j'} h_{k,j'}^j \right) z_j \end{aligned}$$

with Einstein summation convention. Combining with the above we see  $a_k^{N+1} p_{k',\ell}$  is contained in the ideal generated by  $I_k$  and  $\pi$  in  $R[x_1, \dots, x_n, z_1, \dots, z_m]$ . Thus  $p_{k',\ell}$  maps into  $\pi(D_k)_{a_k}$ . On the other hand, the equation

$$\pi p_{k',\ell} = -a_{k'}(f_\ell - \pi z_\ell) + \sum_{j' \in E_{k'}} h_{k',\ell}^{j'}(f_{j'} - \pi z_{j'})$$

shows that  $\pi p_{k',\ell}$  is zero in  $(D_k)_{a_k}$ . Since we have assumed that  $\text{Ann}_R(\pi) = \text{Ann}_R(\pi^2)$  and since  $(D_k)_{q_k}$  is smooth hence flat over  $R$  we see that  $\text{Ann}_{(D_k)_{q_k}}(\pi) = \text{Ann}_{(D_k)_{q_k}}(\pi^2)$ . We conclude that  $p_{k',\ell}$  maps to zero as well, hence  $D_q = (D_k)_{q_k}$  and we win.  $\square$

## 7. The desingularization lemma

Here is another fiendishly clever lemma.

**Lemma 7.1.** *Let  $R$  be a Noetherian ring. Let  $\Lambda$  be an  $R$ -algebra. Let  $\pi \in R$  and assume that  $\text{Ann}_\Lambda(\pi) = \text{Ann}_\Lambda(\pi^2)$ . Let  $A \rightarrow \Lambda$  be an  $R$ -algebra map with  $A$  of finite presentation. Assume*

- (1) *the image of  $\pi$  is strictly standard in  $A$  over  $R$ , and*
- (2) *there exists a section  $\rho : A/\pi^4 A \rightarrow R/\pi^4 R$  which is compatible with the map to  $\Lambda/\pi^4 \Lambda$ .*

*Then we can find  $R$ -algebra maps  $A \rightarrow B \rightarrow \Lambda$  with  $B$  of finite presentation such that  $\mathfrak{a}B \subset H_{B/R}$  where  $\mathfrak{a} = \text{Ann}_R(\text{Ann}_R(\pi^2)/\text{Ann}_R(\pi))$ .*

**Proof.** Choose a presentation

$$A = R[x_1, \dots, x_n]/(f_1, \dots, f_m)$$

and  $0 \leq c \leq \min(n, m)$  such that (2.3.3) holds for  $\pi$  and such that

$$(7.1.1) \quad \pi f_{c+j} \in (f_1, \dots, f_c) + (f_1, \dots, f_m)^2$$

for  $j = 1, \dots, m-c$ . Say  $\rho$  maps  $x_i$  to the class of  $r_i \in R$ . Then we can replace  $x_i$  by  $x_i - r_i$ . Hence we may assume  $\rho(x_i) = 0$  in  $R/\pi^4 R$ . This implies that  $f_j(0) \in \pi^4 R$  and that  $A \rightarrow \Lambda$  maps  $x_i$  to  $\pi^4 \lambda_i$  for some  $\lambda_i \in \Lambda$ . Write

$$f_j = f_j(0) + \sum_{i=1, \dots, n} r_{ji} x_i + \text{h.o.t.}$$

This implies that the constant term of  $\partial f_j / \partial x_i$  is  $r_{ji}$ . Apply  $\rho$  to (2.3.3) for  $\pi$  and we see that

$$\pi = \sum_{I \subset \{1, \dots, n\}, |I|=c} r_I \det(r_{ji})_{j=1, \dots, c, i \in I} \text{ mod } \pi^4 R$$

for some  $r_I \in R$ . Thus we have

$$u\pi = \sum_{I \subset \{1, \dots, n\}, |I|=c} r_I \det(r_{ji})_{j=1, \dots, c, i \in I}$$

for some  $u \in 1 + \pi^3 R$ . By Algebra, Lemma ?? this implies there exists a  $n \times c$  matrix  $(s_{ik})$  such that

$$u\pi \delta_{jk} = \sum_{i=1, \dots, n} r_{ji} s_{ik} \quad \text{for all } j, k = 1, \dots, c$$

(Kronecker delta). We introduce auxiliary variables  $v_1, \dots, v_c, w_1, \dots, w_n$  and we set

$$h_i = x_i - \pi^2 \sum_{j=1, \dots, c} s_{ij} v_j - \pi^3 w_i$$

In the following we will use that

$$R[x_1, \dots, x_n, v_1, \dots, v_c, w_1, \dots, w_n]/(h_1, \dots, h_n) = R[v_1, \dots, v_c, w_1, \dots, w_n]$$

without further mention. In  $R[x_1, \dots, x_n, v_1, \dots, v_c, w_1, \dots, w_n]/(h_1, \dots, h_n)$  we have

$$\begin{aligned} f_j &= f_j(x_1 - h_1, \dots, x_n - h_n) \\ &= \pi^2 \sum_{k=1}^c \left( \sum_{i=1}^n r_{ji} s_{ik} \right) v_k + \pi^3 \sum_{i=1}^n r_{ji} w_i \bmod \pi^4 \\ &= \pi^3 v_j + \pi^3 \sum_{i=1}^n r_{ji} w_i \bmod \pi^4 \end{aligned}$$

for  $1 \leq j \leq c$ . Hence we can choose elements  $g_j \in R[v_1, \dots, v_c, w_1, \dots, w_n]$  such that  $g_j = v_j + \sum r_{ji} w_i \bmod \pi$  and such that  $f_j = \pi^3 g_j$  in the  $R$ -algebra  $R[x_1, \dots, x_n, v_1, \dots, v_c, w_1, \dots, w_n]/(h_1, \dots, h_n)$ . We set

$$B = R[x_1, \dots, x_n, v_1, \dots, v_c, w_1, \dots, w_n]/(f_1, \dots, f_m, h_1, \dots, h_n, g_1, \dots, g_c).$$

The map  $A \rightarrow B$  is clear. We define  $B \rightarrow \Lambda$  by mapping  $x_i \rightarrow \pi^4 \lambda_i$ ,  $v_i \mapsto 0$ , and  $w_i \mapsto \pi \lambda_i$ . Then it is clear that the elements  $f_j$  and  $h_i$  are mapped to zero in  $\Lambda$ . Moreover, it is clear that  $g_i$  is mapped to an element  $t$  of  $\pi \Lambda$  such that  $\pi^3 t = 0$  (as  $f_i = \pi^3 g_i$  modulo the ideal generated by the  $h$ 's). Hence our assumption that  $\text{Ann}_\Lambda(\pi) = \text{Ann}_\Lambda(\pi^2)$  implies that  $t = 0$ . Thus we are done if we can prove the statement about smoothness.

Note that  $B_\pi \cong A_\pi[v_1, \dots, v_c]$  because the equations  $g_i = 0$  are implied by  $f_i = 0$ . Hence  $B_\pi$  is smooth over  $R$  as  $A_\pi$  is smooth over  $R$  by the assumption that  $\pi$  is strictly standard in  $A$  over  $R$ , see Lemma 2.5.

Set  $B' = R[v_1, \dots, v_c, w_1, \dots, w_n]/(g_1, \dots, g_c)$ . As  $g_i = v_i + \sum r_{ji} w_i \bmod \pi$  we see that  $B'/\pi B' = R/\pi R[w_1, \dots, w_n]$ . Hence  $R \rightarrow B'$  is smooth of relative dimension  $n$  at every point of  $V(\pi)$  by Algebra, Lemmas ?? and ?? (the first lemma shows it is syntomic at those primes, in particular flat, whereupon the second lemma shows it is smooth).

Let  $\mathfrak{q} \subset B$  be a prime with  $\pi \in \mathfrak{q}$  and for some  $r \in \mathfrak{a}$ ,  $r \notin \mathfrak{q}$ . Denote  $\mathfrak{q}' = B' \cap \mathfrak{q}$ . We claim the surjection  $B' \rightarrow B$  induces an isomorphism of local rings  $(B')_{\mathfrak{q}'} \rightarrow B_{\mathfrak{q}}$ . This will conclude the proof of the lemma. Note that  $B_{\mathfrak{q}}$  is the quotient of  $(B')_{\mathfrak{q}'}$  by the ideal generated by  $f_{c+j}$ ,  $j = 1, \dots, m - c$ . We observe two things: first the image of  $f_{c+j}$  in  $(B')_{\mathfrak{q}'}$  is divisible by  $\pi^2$  and second the image of  $\pi f_{c+j}$  in  $(B')_{\mathfrak{q}'}$  can be written as  $\sum b_{j_1 j_2} f_{c+j_1} f_{c+j_2}$  by (7.1.1). Thus we see that the image of each  $\pi f_{c+j}$  is contained in the ideal generated by the elements  $\pi^2 f_{c+j'}$ . Hence  $\pi f_{c+j} = 0$  in  $(B')_{\mathfrak{q}'}$  as this is a Noetherian local ring, see Algebra, Lemma ?. As  $R \rightarrow (B')_{\mathfrak{q}'}$  is flat we see that

$$(\text{Ann}_R(\pi^2)/\text{Ann}_R(\pi)) \otimes_R (B')_{\mathfrak{q}'} = \text{Ann}_{(B')_{\mathfrak{q}'}}(\pi^2)/\text{Ann}_{(B')_{\mathfrak{q}'}}(\pi)$$

Because  $r \in \mathfrak{a}$  is invertible in  $(B')_{\mathfrak{q}'}$  we see that this module is zero. Hence we see that the image of  $f_{c+j}$  is zero in  $(B')_{\mathfrak{q}'}$  as desired.  $\square$

**Lemma 7.2.** *Let  $R$  be a Noetherian ring. Let  $\Lambda$  be an  $R$ -algebra. Let  $\pi \in R$  and assume that  $\text{Ann}_R(\pi) = \text{Ann}_R(\pi^2)$  and  $\text{Ann}_\Lambda(\pi) = \text{Ann}_\Lambda(\pi^2)$ . Let  $A \rightarrow \Lambda$  and  $D \rightarrow \Lambda$  be  $R$ -algebra maps with  $A$  and  $D$  of finite presentation. Assume*

- (1)  $\pi$  is strictly standard in  $A$  over  $R$ , and
- (2) *there exists an  $R$ -algebra map  $A/\pi^4 A \rightarrow D/\pi^4 D$  compatible with the maps to  $\Lambda/\pi^4 \Lambda$ .*

Then we can find an  $R$ -algebra map  $B \rightarrow \Lambda$  with  $B$  of finite presentation and  $R$ -algebra maps  $A \rightarrow B$  and  $D \rightarrow B$  compatible with the maps to  $\Lambda$  such that  $H_{D/R}B \subset H_{B/D}$  and  $H_{D/R}B \subset H_{B/R}$ .

**Proof.** We apply Lemma 7.1 to

$$D \longrightarrow A \otimes_R D \longrightarrow \Lambda$$

and the image of  $\pi$  in  $D$ . By Lemma 2.7 we see that  $\pi$  is strictly standard in  $A \otimes_R D$  over  $D$ . As our section  $\rho : (A \otimes_R D)/\pi^4(A \otimes_R D) \rightarrow D/\pi^4D$  we take the map induced by the map in (2). Thus Lemma 7.1 applies and we obtain a factorization  $A \otimes_R D \rightarrow B \rightarrow \Lambda$  with  $B$  of finite presentation and  $\mathfrak{a}B \subset H_{B/D}$  where

$$\mathfrak{a} = \text{Ann}_D(\text{Ann}_D(\pi^2)/\text{Ann}_D(\pi)).$$

For any prime  $\mathfrak{q}$  of  $D$  such that  $D_{\mathfrak{q}}$  is flat over  $R$  we have  $\text{Ann}_{D_{\mathfrak{q}}}(\pi^2)/\text{Ann}_{D_{\mathfrak{q}}}(\pi) = 0$  because annihilators of elements commutes with flat base change and we assumed  $\text{Ann}_R(\pi) = \text{Ann}_R(\pi^2)$ . Because  $D$  is Noetherian we see that  $\text{Ann}_D(\pi^2)/\text{Ann}_D(\pi)$  is a finite  $D$ -module, hence formation of its annihilator commutes with localization. Thus we see that  $\mathfrak{a} \not\subset \mathfrak{q}$ . Hence we see that  $D \rightarrow B$  is smooth at any prime of  $B$  lying over  $\mathfrak{q}$ . Since any prime of  $D$  where  $R \rightarrow D$  is smooth is one where  $D_{\mathfrak{q}}$  is flat over  $R$  we conclude that  $H_{D/R}B \subset H_{B/D}$ . The final inclusion  $H_{D/R}B \subset H_{B/R}$  follows because compositions of smooth ring maps are smooth (Algebra, Lemma ??).  $\square$

**Lemma 7.3.** *Let  $R$  be a Noetherian ring. Let  $\Lambda$  be an  $R$ -algebra. Let  $\pi \in R$  and assume that  $\text{Ann}_R(\pi) = \text{Ann}_R(\pi^2)$  and  $\text{Ann}_{\Lambda}(\pi) = \text{Ann}_{\Lambda}(\pi^2)$ . Let  $A \rightarrow \Lambda$  be an  $R$ -algebra map with  $A$  of finite presentation and assume  $\pi$  is strictly standard in  $A$  over  $R$ . Let*

$$A/\pi^8A \rightarrow \bar{C} \rightarrow \Lambda/\pi^8\Lambda$$

*be a factorization with  $\bar{C}$  of finite presentation. Then we can find a factorization  $A \rightarrow B \rightarrow \Lambda$  with  $B$  of finite presentation such that  $R_{\pi} \rightarrow B_{\pi}$  is smooth and such that*

$$H_{\bar{C}/(R/\pi^8R)} \cdot \Lambda/\pi^8\Lambda \subset \sqrt{H_{B/R}\Lambda} \bmod \pi^8\Lambda.$$

**Proof.** Apply Lemma 6.1 to get  $R \rightarrow D \rightarrow \Lambda$  with a factorization  $\bar{C}/\pi^4\bar{C} \rightarrow D/\pi^4D \rightarrow \Lambda/\pi^4\Lambda$  such that  $R \rightarrow D$  is smooth at any prime not containing  $\pi$  and at any prime lying over a prime of  $\bar{C}/\pi^4\bar{C}$  where  $R/\pi^8R \rightarrow \bar{C}$  is smooth. By Lemma 7.2 we can find a finitely presented  $R$ -algebra  $B$  and factorizations  $A \rightarrow B \rightarrow \Lambda$  and  $D \rightarrow B \rightarrow \Lambda$  such that  $H_{D/R}B \subset H_{B/R}$ . We omit the verification that this is a solution to the problem posed by the lemma.  $\square$

## 8. Warmup: reduction to a base field

In this section we apply the lemmas in the previous sections to prove that it suffices to prove the main result when the base ring is a field, see Lemma 8.4.

**Situation 8.1.** Here  $R \rightarrow \Lambda$  is a regular ring map of Noetherian rings.

Let  $R \rightarrow \Lambda$  be as in Situation 8.1. We say *PT holds for  $R \rightarrow \Lambda$*  if  $\Lambda$  is a filtered colimit of smooth  $R$ -algebras.

**Lemma 8.2.** *Let  $R_i \rightarrow \Lambda_i$ ,  $i = 1, 2$  be as in Situation 8.1. If PT holds for  $R_i \rightarrow \Lambda_i$ ,  $i = 1, 2$ , then PT holds for  $R_1 \times R_2 \rightarrow \Lambda_1 \times \Lambda_2$ .*

**Proof.** Omitted. Hint: A product of filtered colimits is a filtered colimit.  $\square$

**Lemma 8.3.** *Let  $R \rightarrow A \rightarrow \Lambda$  be ring maps with  $A$  of finite presentation over  $R$ . Let  $S \subset R$  be a multiplicative set. Let  $S^{-1}A \rightarrow B' \rightarrow S^{-1}\Lambda$  be a factorization with  $B'$  smooth over  $S^{-1}R$ . Then we can find a factorization  $A \rightarrow B \rightarrow \Lambda$  such that some  $s \in S$  maps to an elementary standard element (Definition 2.3) in  $B$  over  $R$ .*

**Proof.** We first apply Lemma 3.4 to  $S^{-1}R \rightarrow B'$ . Thus we may assume  $B'$  is standard smooth over  $S^{-1}R$ . Write  $A = R[x_1, \dots, x_n]/(g_1, \dots, g_t)$  and say  $x_i \mapsto \lambda_i$  in  $\Lambda$ . We may write  $B' = S^{-1}R[x_1, \dots, x_{n+m}]/(f_1, \dots, f_c)$  for some  $c \geq n$  where  $\det(\partial f_j / \partial x_i)_{i,j=1, \dots, c}$  is invertible in  $B'$  and such that  $A \rightarrow B'$  is given by  $x_i \mapsto x_i$ , see Lemma 3.6. After multiplying  $x_i$ ,  $i > n$  by an element of  $S$  and correspondingly modifying the equations  $f_j$  we may assume  $B' \rightarrow S^{-1}\Lambda$  maps  $x_i$  to  $\lambda_i/1$  for some  $\lambda_i \in \Lambda$  for  $i > n$ . Choose a relation

$$1 = a_0 \det(\partial f_j / \partial x_i)_{i,j=1, \dots, c} + \sum_{j=1, \dots, c} a_j f_j$$

for some  $a_j \in S^{-1}R[x_1, \dots, x_{n+m}]$ . Since each element of  $S$  is invertible in  $B'$  we may (by clearing denominators) assume that  $f_j, a_j \in R[x_1, \dots, x_{n+m}]$  and that

$$s_0 = a_0 \det(\partial f_j / \partial x_i)_{i,j=1, \dots, c} + \sum_{j=1, \dots, c} a_j f_j$$

for some  $s_0 \in S$ . Since  $g_j$  maps to zero in  $S^{-1}R[x_1, \dots, x_{n+m}]/(f_1, \dots, f_c)$  we can find elements  $s_j \in S$  such that  $s_j g_j = 0$  in  $R[x_1, \dots, x_{n+m}]/(f_1, \dots, f_c)$ . Since  $f_j$  maps to zero in  $S^{-1}\Lambda$  we can find  $s'_j \in S$  such that  $s'_j f_j(\lambda_1, \dots, \lambda_{n+m}) = 0$  in  $\Lambda$ . Consider the ring

$$B = R[x_1, \dots, x_{n+m}]/(s'_1 f_1, \dots, s'_c f_c, g_1, \dots, g_t)$$

and the factorization  $A \rightarrow B \rightarrow \Lambda$  with  $B \rightarrow \Lambda$  given by  $x_i \mapsto \lambda_i$ . We claim that  $s = s_0 s_1 \dots s_t s'_1 \dots s'_c$  is elementary standard in  $B$  over  $R$  which finishes the proof. Namely,  $s_j g_j \in (f_1, \dots, f_c)$  and hence  $sg_j \in (s'_1 f_1, \dots, s'_c f_c)$ . Finally, we have

$$a_0 \det(\partial s'_j f_j / \partial x_i)_{i,j=1, \dots, c} + \sum_{j=1, \dots, c} (s'_1 \dots \hat{s}'_j \dots s'_c) a_j s'_j f_j = s_0 s'_1 \dots s'_c$$

which divides  $s$  as desired.  $\square$

**Lemma 8.4.** *If for every Situation 8.1 where  $R$  is a field PT holds, then PT holds in general.*

**Proof.** Assume PT holds for any Situation 8.1 where  $R$  is a field. Let  $R \rightarrow \Lambda$  be as in Situation 8.1 arbitrary. Note that  $R/I \rightarrow \Lambda/I\Lambda$  is another regular ring map of Noetherian rings, see More on Algebra, Lemma ???. Consider the set of ideals

$$\mathcal{I} = \{I \subset R \mid R/I \rightarrow \Lambda/I\Lambda \text{ does not have PT}\}$$

We have to show that  $\mathcal{I}$  is empty. If this set is nonempty, then it contains a maximal element because  $R$  is Noetherian. Replacing  $R$  by  $R/I$  and  $\Lambda$  by  $\Lambda/I$  we obtain a situation where PT holds for  $R/I \rightarrow \Lambda/I\Lambda$  for any nonzero ideal of  $R$ . In particular, we see by applying Proposition 5.3 that  $R$  is a reduced ring.

Let  $A \rightarrow \Lambda$  be an  $R$ -algebra homomorphism with  $A$  of finite presentation. We have to find a factorization  $A \rightarrow B \rightarrow \Lambda$  with  $B$  smooth over  $R$ , see Algebra, Lemma ???.

Let  $S \subset R$  be the set of nonzerodivisors and consider the total ring of fractions  $Q = S^{-1}R$  of  $R$ . We know that  $Q = K_1 \times \dots \times K_n$  is a product of fields, see

Algebra, Lemmas ?? and ?. By Lemma 8.2 and our assumption PT holds for the ring map  $S^{-1}R \rightarrow S^{-1}\Lambda$ . Hence we can find a factorization  $S^{-1}A \rightarrow B' \rightarrow S^{-1}\Lambda$  with  $B'$  smooth over  $S^{-1}R$ .

We apply Lemma 8.3 and find a factorization  $A \rightarrow B \rightarrow \Lambda$  such that some  $\pi \in S$  is elementary standard in  $B$  over  $R$ . After replacing  $A$  by  $B$  we may assume that  $\pi$  is elementary standard, hence strictly standard in  $A$ . We know that  $R/\pi^8 R \rightarrow \Lambda/\pi^8 \Lambda$  satisfies PT. Hence we can find a factorization  $R/\pi^8 R \rightarrow A/\pi^8 A \rightarrow \bar{C} \rightarrow \Lambda/\pi^8 \Lambda$  with  $R/\pi^8 R \rightarrow \bar{C}$  smooth. By Lemma 6.1 we can find an  $R$ -algebra map  $D \rightarrow \Lambda$  with  $D$  smooth over  $R$  and a factorization  $R/\pi^4 R \rightarrow A/\pi^4 A \rightarrow D/\pi^4 D \rightarrow \Lambda/\pi^4 \Lambda$ . By Lemma 7.2 we can find  $A \rightarrow B \rightarrow \Lambda$  with  $B$  smooth over  $R$  which finishes the proof.  $\square$

## 9. Local tricks

**Situation 9.1.** We are given a Noetherian ring  $R$  and an  $R$ -algebra map  $A \rightarrow \Lambda$  and a prime  $\mathfrak{q} \subset \Lambda$ . We assume  $A$  is of finite presentation over  $R$ . In this situation we denote  $\mathfrak{h}_A = \sqrt{H_{A/R}\Lambda}$ .

Let  $R \rightarrow A \rightarrow \Lambda \supset \mathfrak{q}$  be as in Situation 9.1. We say  $R \rightarrow A \rightarrow \Lambda \supset \mathfrak{q}$  can be resolved if there exists a factorization  $A \rightarrow B \rightarrow \Lambda$  with  $B$  of finite presentation and  $\mathfrak{h}_A \subset \mathfrak{h}_B \not\subset \mathfrak{q}$ . In this case we will call the factorization  $A \rightarrow B \rightarrow \Lambda$  a resolution of  $R \rightarrow A \rightarrow \Lambda \supset \mathfrak{q}$ .

**Lemma 9.2.** Let  $R \rightarrow A \rightarrow \Lambda \supset \mathfrak{q}$  be as in Situation 9.1. Let  $r \geq 1$  and  $\pi_1, \dots, \pi_r \in R$  map to elements of  $\mathfrak{q}$ . Assume

- (1) for  $i = 1, \dots, r$  we have

$$\text{Ann}_{R/(\pi_1^8, \dots, \pi_{i-1}^8)R}(\pi_i) = \text{Ann}_{R/(\pi_1^8, \dots, \pi_{i-1}^8)R}(\pi_i^2)$$

and

$$\text{Ann}_{\Lambda/(\pi_1^8, \dots, \pi_{i-1}^8)\Lambda}(\pi_i) = \text{Ann}_{\Lambda/(\pi_1^8, \dots, \pi_{i-1}^8)\Lambda}(\pi_i^2)$$

- (2) for  $i = 1, \dots, r$  the element  $\pi_i$  maps to a strictly standard element in  $A$  over  $R$ .

Then, if

$$R/(\pi_1^8, \dots, \pi_r^8)R \rightarrow A/(\pi_1^8, \dots, \pi_r^8)A \rightarrow \Lambda/(\pi_1^8, \dots, \pi_r^8)\Lambda \supset \mathfrak{q}/(\pi_1^8, \dots, \pi_r^8)\Lambda$$

can be resolved, so can  $R \rightarrow A \rightarrow \Lambda \supset \mathfrak{q}$ .

**Proof.** We are going to prove this by induction on  $r$ .

The case  $r = 1$ . Here the assumption is that there exists a factorization  $A/\pi_1^8 \rightarrow \bar{C} \rightarrow \Lambda/\pi_1^8$  which resolves the situation modulo  $\pi_1^8$ . Conditions (1) and (2) are the assumptions needed to apply Lemma 7.3. Thus we can “lift” the resolution  $\bar{C}$  to a resolution of  $R \rightarrow A \rightarrow \Lambda \supset \mathfrak{q}$ .

The case  $r > 1$ . In this case we apply the induction hypothesis for  $r - 1$  to the situation  $R/\pi_1^8 \rightarrow A/\pi_1^8 \rightarrow \Lambda/\pi_1^8 \supset \mathfrak{q}/\pi_1^8 \Lambda$ . Note that property (2) is preserved by Lemma 2.7.  $\square$

**Lemma 9.3.** Let  $R \rightarrow A \rightarrow \Lambda \supset \mathfrak{q}$  be as in Situation 9.1. Let  $\mathfrak{p} = R \cap \mathfrak{q}$ . Assume that  $\mathfrak{q}$  is minimal over  $\mathfrak{h}_A$  and that  $R_{\mathfrak{p}} \rightarrow A_{\mathfrak{p}} \rightarrow \Lambda_{\mathfrak{q}} \supset \mathfrak{q}\Lambda_{\mathfrak{q}}$  can be resolved. Then there exists a factorization  $A \rightarrow C \rightarrow \Lambda$  with  $C$  of finite presentation such that  $H_{C/R}\Lambda \not\subset \mathfrak{q}$ .

**Proof.** Let  $A_{\mathfrak{p}} \rightarrow C \rightarrow \Lambda_{\mathfrak{q}}$  be a resolution of  $R_{\mathfrak{p}} \rightarrow A_{\mathfrak{p}} \rightarrow \Lambda_{\mathfrak{q}} \supset \mathfrak{q}\Lambda_{\mathfrak{q}}$ . By our assumption that  $\mathfrak{q}$  is minimal over  $\mathfrak{h}_A$  this means that  $H_{C/R_{\mathfrak{p}}}\Lambda_{\mathfrak{q}} = \Lambda_{\mathfrak{q}}$ . By Lemma 2.8 we may assume that  $C$  is smooth over  $R_{\mathfrak{p}}$ . By Lemma 3.4 we may assume that  $C$  is standard smooth over  $R_{\mathfrak{p}}$ . Write  $A = R[x_1, \dots, x_n]/(g_1, \dots, g_t)$  and say  $A \rightarrow \Lambda$  is given by  $x_i \mapsto \lambda_i$ . Write  $C = R_{\mathfrak{p}}[x_1, \dots, x_{n+m}]/(f_1, \dots, f_c)$  for some  $c \geq n$  such that  $A \rightarrow C$  maps  $x_i$  to  $x_i$  and such that  $\det(\partial f_j / \partial x_i)_{i,j=1,\dots,c}$  is invertible in  $C$ , see Lemma 3.6. After clearing denominators we may assume  $f_1, \dots, f_c$  are elements of  $R[x_1, \dots, x_{n+m}]$ . Of course  $\det(\partial f_j / \partial x_i)_{i,j=1,\dots,c}$  is not invertible in  $R[x_1, \dots, x_{n+m}]/(f_1, \dots, f_c)$  but it becomes invertible after inverting some element  $s_0 \in R$ ,  $s_0 \notin \mathfrak{p}$ . As  $g_j$  maps to zero under  $R[x_1, \dots, x_n] \rightarrow A \rightarrow C$  we can find  $s_j \in R$ ,  $s_j \notin \mathfrak{p}$  such that  $s_j g_j$  is zero in  $R[x_1, \dots, x_{n+m}]/(f_1, \dots, f_c)$ . Write  $f_j = F_j(x_1, \dots, x_{n+m}, 1)$  for some polynomial  $F_j \in R[x_1, \dots, x_n, X_{n+1}, \dots, X_{n+m+1}]$  homogeneous in  $X_{n+1}, \dots, X_{n+m+1}$ . Pick  $\lambda_{n+i} \in \Lambda$ ,  $i = 1, \dots, m+1$  with  $\lambda_{n+m+1} \notin \mathfrak{q}$  such that  $x_{n+i}$  maps to  $\lambda_{n+i}/\lambda_{n+m+1}$  in  $\Lambda_{\mathfrak{q}}$ . Then

$$\begin{aligned} F_j(\lambda_1, \dots, \lambda_{n+m+1}) &= (\lambda_{n+m+1})^{\deg(F_j)} F_j(\lambda_1, \dots, \lambda_n, \frac{\lambda_{n+1}}{\lambda_{n+m+1}}, \dots, \frac{\lambda_{n+m}}{\lambda_{n+m+1}}, 1) \\ &= (\lambda_{n+m+1})^{\deg(F_j)} f_j(\lambda_1, \dots, \lambda_n, \frac{\lambda_{n+1}}{\lambda_{n+m+1}}, \dots, \frac{\lambda_{n+m}}{\lambda_{n+m+1}}) \\ &= 0 \end{aligned}$$

in  $\Lambda_{\mathfrak{q}}$ . Thus we can find  $\lambda_0 \in \Lambda$ ,  $\lambda_0 \notin \mathfrak{q}$  such that  $\lambda_0 F_j(\lambda_1, \dots, \lambda_{n+m+1}) = 0$  in  $\Lambda$ . Now we set  $B$  equal to

$$R[x_0, \dots, x_{n+m+1}]/(g_1, \dots, g_t, x_0 F_1(x_1, \dots, x_{n+m+1}), \dots, x_0 F_c(x_1, \dots, x_{n+m+1}))$$

which we map to  $\Lambda$  by mapping  $x_i$  to  $\lambda_i$ . Let  $b$  be the image of  $x_0 x_{n+m+1} s_0 s_1 \dots s_t$  in  $B$ . Then  $B_b$  is isomorphic to

$$R_{s_0 s_1 \dots s_t}[x_0, x_1, \dots, x_{n+m+1}, 1/x_0 x_{n+m+1}]/(f_1, \dots, f_c)$$

which is smooth over  $R$  by construction. Since  $b$  does not map to an element of  $\mathfrak{q}$ , we win.  $\square$

**Lemma 9.4.** *Let  $R \rightarrow A \rightarrow \Lambda \supset \mathfrak{q}$  be as in Situation 9.1. Let  $\mathfrak{p} = R \cap \mathfrak{q}$ . Assume*

- (1)  $\mathfrak{q}$  is minimal over  $\mathfrak{h}_A$ ,
- (2)  $R_{\mathfrak{p}} \rightarrow A_{\mathfrak{p}} \rightarrow \Lambda_{\mathfrak{q}} \supset \mathfrak{q}\Lambda_{\mathfrak{q}}$  can be resolved, and
- (3)  $\dim(\Lambda_{\mathfrak{q}}) = 0$ .

*Then  $R \rightarrow A \rightarrow \Lambda \supset \mathfrak{q}$  can be resolved.*

**Proof.** By (3) the ring  $\Lambda_{\mathfrak{q}}$  is Artinian local hence  $\mathfrak{q}\Lambda_{\mathfrak{q}}$  is nilpotent. Thus  $(\mathfrak{h}_A)^N \Lambda_{\mathfrak{q}} = 0$  for some  $N > 0$ . Thus there exists a  $\lambda \in \Lambda$ ,  $\lambda \notin \mathfrak{q}$  such that  $\lambda(\mathfrak{h}_A)^N = 0$  in  $\Lambda$ . Say  $H_{A/R} = (a_1, \dots, a_r)$  so that  $\lambda a_i^N = 0$  in  $\Lambda$ . By Lemma 9.3 we can find a factorization  $A \rightarrow C \rightarrow \Lambda$  with  $C$  of finite presentation such that  $\mathfrak{h}_C \not\subset \mathfrak{q}$ . Write  $C = A[x_1, \dots, x_n]/(f_1, \dots, f_m)$ . Set

$$B = A[x_1, \dots, x_n, y_1, \dots, y_r, z, t_{ij}]/(f_j - \sum y_i t_{ij}, z y_i)$$

where  $t_{ij}$  is a set of  $rm$  variables. Note that there is a map  $B \rightarrow C[y_i, z]/(y_i z)$  given by setting  $t_{ij}$  equal to zero. The map  $B \rightarrow \Lambda$  is the composition  $B \rightarrow C[y_i, z]/(y_i z) \rightarrow \Lambda$  where  $C[y_i, z]/(y_i z) \rightarrow \Lambda$  is the given map  $C \rightarrow \Lambda$ , maps  $z$  to  $\lambda$ , and maps  $y_i$  to the image of  $a_i^N$  in  $\Lambda$ .

We claim that  $B$  is a solution for  $R \rightarrow A \rightarrow \Lambda \supset \mathfrak{q}$ . First note that  $B_z$  is isomorphic to  $C[y_1, \dots, y_r, z, z^{-1}]$  and hence is smooth. On the other hand,  $B_{y_\ell} \cong A[x_i, y_i, y_\ell^{-1}, t_{ij}, i \neq \ell]$  which is smooth over  $A$ . Thus we see that  $z$  and  $a_\ell y_\ell$  (compositions of smooth maps are smooth) are all elements of  $H_{B/R}$ . This proves the lemma.  $\square$

## 10. Separable residue fields

In this section we explain how to solve a local problem in the case of a separable residue field extension.

**Lemma 10.1** (Ogoma). *Let  $A$  be a Noetherian ring and let  $M$  be a finite  $A$ -module. Let  $S \subset A$  be a multiplicative set. If  $\pi \in A$  and  $\text{Ker}(\pi : S^{-1}M \rightarrow S^{-1}M) = \text{Ker}(\pi^2 : S^{-1}M \rightarrow S^{-1}M)$  then there exists an  $s \in S$  such that for any  $n > 0$  we have  $\text{Ker}(s^n \pi : M \rightarrow M) = \text{Ker}((s^n \pi)^2 : M \rightarrow M)$ .*

**Proof.** Let  $K = \text{Ker}(\pi : M \rightarrow M)$  and  $K' = \{m \in M \mid \pi^2 m = 0 \text{ in } S^{-1}M\}$  and  $Q = K'/K$ . Note that  $S^{-1}Q = 0$  by assumption. Since  $A$  is Noetherian we see that  $Q$  is a finite  $A$ -module. Hence we can find an  $s \in S$  such that  $s$  annihilates  $Q$ . Then  $s$  works.  $\square$

**Lemma 10.2.** *Let  $\Lambda$  be a Noetherian ring. Let  $I \subset \Lambda$  be an ideal. Let  $I \subset \mathfrak{q}$  be a prime. Let  $n, e$  be positive integers. Assume that  $\mathfrak{q}^n \Lambda_{\mathfrak{q}} \subset I \Lambda_{\mathfrak{q}}$  and that  $\Lambda_{\mathfrak{q}}$  is a regular local ring of dimension  $d$ . Then there exists an  $n > 0$  and  $\pi_1, \dots, \pi_d \in \Lambda$  such that*

- (1)  $(\pi_1, \dots, \pi_d) \Lambda_{\mathfrak{q}} = \mathfrak{q} \Lambda_{\mathfrak{q}}$ ,
- (2)  $\pi_1^n, \dots, \pi_d^n \in I$ , and
- (3) for  $i = 1, \dots, d$  we have

$$\text{Ann}_{\Lambda/(\pi_1^e, \dots, \pi_{i-1}^e) \Lambda}(\pi_i) = \text{Ann}_{\Lambda/(\pi_1^e, \dots, \pi_{i-1}^e) \Lambda}(\pi_i^2).$$

**Proof.** Set  $S = \Lambda \setminus \mathfrak{q}$  so that  $\Lambda_{\mathfrak{q}} = S^{-1}\Lambda$ . First pick  $\pi_1, \dots, \pi_d$  with (1) which is possible as  $\Lambda_{\mathfrak{q}}$  is regular. By assumption  $\pi_i^n \in I \Lambda_{\mathfrak{q}}$ . Thus we can find  $s_1, \dots, s_d \in S$  such that  $s_i \pi_i^n \in I$ . Replacing  $\pi_i$  by  $s_i \pi_i$  we get (2). Note that (1) and (2) are preserved by further multiplying by elements of  $S$ . Suppose that (3) holds for  $i = 1, \dots, t$  for some  $t \in \{0, \dots, d\}$ . Note that  $\pi_1, \dots, \pi_d$  is a regular sequence in  $S^{-1}\Lambda$ , see Algebra, Lemma ???. In particular  $\pi_1^e, \dots, \pi_t^e, \pi_{t+1}$  is a regular sequence in  $S^{-1}\Lambda = \Lambda_{\mathfrak{q}}$  by Algebra, Lemma ???. Hence we see that

$$\text{Ann}_{S^{-1}\Lambda/(\pi_1^e, \dots, \pi_{i-1}^e) \Lambda}(\pi_i) = \text{Ann}_{S^{-1}\Lambda/(\pi_1^e, \dots, \pi_{i-1}^e) \Lambda}(\pi_i^2).$$

Thus we get (3) for  $i = t+1$  after replacing  $\pi_{t+1}$  by  $s \pi_{t+1}$  for some  $s \in S$  by Lemma 10.1. By induction on  $t$  this produces a sequence satisfying (1), (2), and (3).  $\square$

**Lemma 10.3.** *Let  $k \rightarrow A \rightarrow \Lambda \supset \mathfrak{q}$  be as in Situation 9.1 where*

- (1)  $k$  is a field,
- (2)  $\Lambda$  is Noetherian,
- (3)  $\mathfrak{q}$  is minimal over  $\mathfrak{h}_A$ ,
- (4)  $\Lambda_{\mathfrak{q}}$  is a regular local ring, and
- (5) the field extension  $\kappa(\mathfrak{q})/k$  is separable.

*Then  $k \rightarrow A \rightarrow \Lambda \supset \mathfrak{q}$  can be resolved.*



**Proof.** Set  $d = \dim \Lambda_{\mathfrak{q}}$ . Set  $R = k[x_1, \dots, x_d]$ . Choose  $n > 0$  such that  $\mathfrak{q}^n \Lambda_{\mathfrak{q}} \subset \mathfrak{h}_A \Lambda_{\mathfrak{q}}$  which is possible as  $\mathfrak{q}$  is minimal over  $\mathfrak{h}_A$ . Choose generators  $a_1, \dots, a_r$  of  $H_{A/R}$ . Set

$$B = A[x_1, \dots, x_d, z_{ij}]/(x_i^n - \sum z_{ij} a_j)$$

Each  $B_{a_j}$  is smooth over  $R$  it is a polynomial algebra over  $A_{a_j}[x_1, \dots, x_d]$  and  $A_{a_j}$  is smooth over  $k$ . Hence  $B_{x_i}$  is smooth over  $R$ . Let  $B \rightarrow C$  be the  $R$ -algebra map constructed in Lemma 3.1 which comes with a  $R$ -algebra retraction  $C \rightarrow B$ . In particular a map  $C \rightarrow \Lambda$  fitting into the diagram above. By construction  $C_{x_i}$  is a smooth  $R$ -algebra with  $\Omega_{C_{x_i}/R}$  free. Hence we can find  $c > 0$  such that  $x_i^c$  is strictly standard in  $C/R$ , see Lemma 3.7. Now choose  $\pi_1, \dots, \pi_d \in \Lambda$  as in Lemma 10.2 where  $n = n$ ,  $e = 8c$ ,  $\mathfrak{q} = \mathfrak{q}$  and  $I = \mathfrak{h}_A$ . Write  $\pi_i^n = \sum \lambda_{ij} a_j$  for some  $\pi_{ij} \in \Lambda$ . There is a map  $B \rightarrow \Lambda$  given by  $x_i \mapsto \pi_i$  and  $z_{ij} \mapsto \lambda_{ij}$ . Set  $R = k[x_1, \dots, x_d]$ . Diagram

$$\begin{array}{ccccc} R & \longrightarrow & B & & \\ \uparrow & & \uparrow & \searrow & \\ k & \longrightarrow & A & \longrightarrow & \Lambda \end{array}$$

Now we apply Lemma 9.2 to  $R \rightarrow C \rightarrow \Lambda \supset \mathfrak{q}$  and the sequence of elements  $x_1^c, \dots, x_d^c$  of  $R$ . Assumption (2) is clear. Assumption (1) holds for  $R$  by inspection and for  $\Lambda$  by our choice of  $\pi_1, \dots, \pi_d$ . (Note that if  $\text{Ann}_{\Lambda}(\pi) = \text{Ann}_{\Lambda}(\pi^2)$ , then we have  $\text{Ann}_{\Lambda}(\pi) = \text{Ann}_{\Lambda}(\pi^c)$  for all  $c > 0$ .) Thus it suffices to resolve

$$R/(x_1^e, \dots, x_d^e) \rightarrow C/(x_1^e, \dots, x_d^e) \rightarrow \Lambda/(\pi_1^e, \dots, \pi_d^e) \supset \mathfrak{q}/(\pi_1^e, \dots, \pi_d^e)$$

for  $e = 8c$ . By Lemma 9.4 it suffices to resolve this after localizing at  $\mathfrak{q}$ . But since  $x_1, \dots, x_d$  map to a regular sequence in  $\Lambda_{\mathfrak{q}}$  we see that  $R_{\mathfrak{p}} \rightarrow \Lambda_{\mathfrak{q}}$  is flat, see Algebra, Lemma ???. Hence

$$R_{\mathfrak{p}}/(x_1^e, \dots, x_d^e) \rightarrow \Lambda_{\mathfrak{q}}/(\pi_1^e, \dots, \pi_d^e)$$

is a flat ring map of Artinian local rings. Moreover, this map induces a separable field extension on residue fields by assumption. Thus this map is a filtered colimit of smooth algebras by Algebra, Lemma ?? and Proposition 5.3. Existence of the desired solution follows from Algebra, Lemma ???.  $\square$

## 11. Inseparable residue fields

In this section we explain how to solve a local problem in the case of an inseparable residue field extension.

**Lemma 11.1.** *Let  $k$  be a field of characteristic  $p > 0$ . Let  $(\Lambda, \mathfrak{m}, K)$  be an Artinian local  $k$ -algebra. Assume that  $\dim H_1(L_{K/k}) < \infty$ . Then  $\Lambda$  is a filtered colimit of Artinian local  $k$ -algebras  $A$  with each map  $A \rightarrow \Lambda$  flat, with  $\mathfrak{m}_A \Lambda = \mathfrak{m}$ , and with  $A$  essentially of finite type over  $k$ .*

**Proof.** Note that the flatness of  $A \rightarrow \Lambda$  implies that  $A \rightarrow \Lambda$  is injective, so the lemma really tells us that  $\Lambda$  is a directed union of these types of subrings  $A \subset \Lambda$ . Let  $n$  be the minimal integer such that  $\mathfrak{m}^n = 0$ . We will prove this lemma by induction on  $n$ . The case  $n = 1$  is clear as a field extension is a union of finitely generated field extensions.

Pick  $\lambda_1, \dots, \lambda_d \in \mathfrak{m}$  which generate  $\mathfrak{m}$ . As  $K$  is formally smooth over  $\mathbf{F}_p$  (see Algebra, Lemma ??) we can find a ring map  $\sigma : K \rightarrow \Lambda$  which is a section of the quotient map  $\Lambda \rightarrow K$ . In general  $\sigma$  is **not** a  $k$ -algebra map. Given  $\sigma$  we define

$$\Psi_\sigma : K[x_1, \dots, x_d] \longrightarrow \Lambda$$

using  $\sigma$  on elements of  $K$  and mapping  $x_i$  to  $\lambda_i$ . Claim: there exists a  $\sigma : K \rightarrow \Lambda$  and a subfield  $k \subset F \subset K$  finitely generated over  $k$  such that the image of  $k$  in  $\Lambda$  is contained in  $\Psi_\sigma(F[x_1, \dots, x_d])$ .

We will prove the claim by induction on the least integer  $n$  such that  $\mathfrak{m}^n = 0$ . It is clear for  $n = 1$ . If  $n > 1$  set  $I = \mathfrak{m}^{n-1}$  and  $\Lambda' = \Lambda/I$ . By induction we may assume given  $\sigma' : K \rightarrow \Lambda'$  and  $k \subset F' \subset K$  finitely generated such that the image of  $k \rightarrow \Lambda \rightarrow \Lambda'$  is contained in  $A' = \Psi_{\sigma'}(F'[x_1, \dots, x_d])$ . Denote  $\tau' : k \rightarrow A'$  the induced map. Choose a lift  $\sigma : K \rightarrow \Lambda$  of  $\sigma'$  (this is possible by the formal smoothness of  $K/\mathbf{F}_p$  we mentioned above). For later reference we note that we can change  $\sigma$  to  $\sigma + D$  for some derivation  $D : K \rightarrow I$ . Set  $A = F[x_1, \dots, x_d]/(x_1, \dots, x_d)^n$ . Then  $\Psi_\sigma$  induces a ring map  $\Psi_\sigma : A \rightarrow \Lambda$ . The composition with the quotient map  $\Lambda \rightarrow \Lambda'$  induces a surjective map  $A \rightarrow A'$  with nilpotent kernel. Choose a lift  $\tau : k \rightarrow A$  of  $\tau'$  (possible as  $k/\mathbf{F}_p$  is formally smooth). Thus we obtain two maps  $k \rightarrow \Lambda$ , namely  $\Psi_\sigma \circ \tau : k \rightarrow \Lambda$  and the given map  $i : k \rightarrow \Lambda$ . These maps agree modulo  $I$ , whence the difference is a derivation  $\theta = i - \Psi_\sigma \circ \tau : k \rightarrow I$ . Note that if we change  $\sigma$  into  $\sigma + D$  then we change  $\theta$  into  $\theta - D|_k$ .

Choose a set of elements  $\{y_j\}_{j \in J}$  of  $k$  whose differentials  $dy_j$  form a basis of  $\Omega_{k/\mathbf{F}_p}$ . The Jacobi-Zariski sequence for  $\mathbf{F}_p \subset k \subset K$  is

$$0 \rightarrow H_1(L_{K/k}) \rightarrow \Omega_{k/\mathbf{F}_p} \otimes K \rightarrow \Omega_{K/\mathbf{F}_p} \rightarrow \Omega_{K/k} \rightarrow 0$$

As  $\dim H_1(L_{K/k}) < \infty$  we can find a finite subset  $J_0 \subset J$  such that the image of the first map is contained in  $\bigoplus_{j \in J_0} K dy_j$ . Hence the elements  $dy_j$ ,  $j \in J \setminus J_0$  map to  $K$ -linearly independent elements of  $\Omega_{K/\mathbf{F}_p}$ . Therefore we can choose a  $D : K \rightarrow I$  such that  $\theta - D|_k = \xi \circ d$  where  $\xi$  is a composition

$$\Omega_{k/\mathbf{F}_p} = \bigoplus_{j \in J} k dy_j \longrightarrow \bigoplus_{j \in J_0} k dy_j \longrightarrow I$$

Let  $f_j = \xi(dy_j) \in I$  for  $j \in J_0$ . Change  $\sigma$  into  $\sigma + D$  as above. Then we see that  $\theta(a) = \sum_{j \in J_0} a_j f_j$  for  $a \in k$  where  $da = \sum a_j dy_j$  in  $\Omega_{k/\mathbf{F}_p}$ . Note that  $I$  is generated by the monomials  $\lambda^E = \lambda_1^{e_1} \dots \lambda_d^{e_d}$  of total degree  $|E| = \sum e_i = n - 1$  in  $\lambda_1, \dots, \lambda_d$ . Write  $f_j = \sum_E c_{j,E} \lambda^E$  with  $c_{j,E} \in K$ . Replace  $F'$  by  $F = F'(c_{j,E})$ . Then the claim holds.

Choose  $\sigma$  and  $F$  as in the claim. The kernel of  $\Psi_\sigma$  is generated by finitely many polynomials  $g_1, \dots, g_t \in K[x_1, \dots, x_d]$  and we may assume their coefficients are in  $F$  after enlarging  $F$  by adjoining finitely many elements. In this case it is clear that the map  $A = F[x_1, \dots, x_d]/(g_1, \dots, g_t) \rightarrow K[x_1, \dots, x_d]/(g_1, \dots, g_t) = \Lambda$  is flat. By the claim  $A$  is a  $k$ -subalgebra of  $\Lambda$ . It is clear that  $\Lambda$  is the filtered colimit of these algebras, as  $K$  is the filtered union of the subfields  $F$ . Finally, these algebras are essentially of finite type over  $k$  by Algebra, Lemma ??.

**Lemma 11.2.** *Let  $k$  be a field of characteristic  $p > 0$ . Let  $\Lambda$  be a Noetherian geometrically regular  $k$ -algebra. Let  $\mathfrak{q} \subset \Lambda$  be a prime ideal. Let  $n \geq 1$  be an integer and let  $E \subset \Lambda_{\mathfrak{q}}/\mathfrak{q}^n \Lambda_{\mathfrak{q}}$  be a finite subset. Then we can find  $m \geq 0$  and  $\varphi : k[y_1, \dots, y_m] \rightarrow \Lambda$  with the following properties*

- (1) setting  $\mathfrak{p} = \varphi^{-1}(\mathfrak{q})$  we have  $\mathfrak{q}\Lambda_{\mathfrak{q}} = \mathfrak{p}\Lambda_{\mathfrak{q}}$  and  $k[y_1, \dots, y_m]_{\mathfrak{p}} \rightarrow \Lambda_{\mathfrak{q}}$  is flat,
- (2) there is a factorization by homomorphisms of local Artinian rings

$$k[y_1, \dots, y_m]_{\mathfrak{p}}/\mathfrak{p}^n k[y_1, \dots, y_m]_{\mathfrak{p}} \rightarrow D \rightarrow \Lambda_{\mathfrak{q}}/\mathfrak{q}^n \Lambda_{\mathfrak{q}}$$

where the first arrow is essentially smooth and the second is flat,

- (3)  $E$  is contained in  $D$  modulo  $\mathfrak{q}^n \Lambda_{\mathfrak{q}}$ .

**Proof.** Set  $\bar{\Lambda} = \Lambda_{\mathfrak{q}}/\mathfrak{q}^n \Lambda_{\mathfrak{q}}$ . Note that  $\dim H_1(L_{\kappa(\mathfrak{q})/k}) < \infty$  by More on Algebra, Proposition ???. Pick  $A \subset \bar{\Lambda}$  containing  $E$  such that  $A$  is local Artinian, essentially of finite type over  $k$ , the map  $A \rightarrow \bar{\Lambda}$  is flat, and  $\mathfrak{m}_A$  generates the maximal ideal of  $\bar{\Lambda}$ , see Lemma 11.1. Denote  $F = A/\mathfrak{m}_A$  the residue field so that  $k \subset F \subset K$ . Pick  $\lambda_1, \dots, \lambda_t \in \Lambda$  which map to elements of  $A$  in  $\bar{\Lambda}$  such that moreover the images of  $d\lambda_1, \dots, d\lambda_t$  form a basis of  $\Omega_{F/k}$ . Consider the map  $\varphi' : k[y_1, \dots, y_t] \rightarrow \Lambda$  sending  $y_j$  to  $\lambda_j$ . Set  $\mathfrak{p}' = (\varphi')^{-1}(\mathfrak{q})$ . By More on Algebra, Lemma ?? the ring map  $k[y_1, \dots, y_t]_{\mathfrak{p}'} \rightarrow \Lambda_{\mathfrak{q}}$  is flat and  $\Lambda_{\mathfrak{q}}/\mathfrak{p}'\Lambda_{\mathfrak{q}}$  is regular. Thus we can choose further elements  $\lambda_{t+1}, \dots, \lambda_m \in \Lambda$  which map into  $A \subset \bar{\Lambda}$  and which map to a regular system of parameters of  $\Lambda_{\mathfrak{q}}/\mathfrak{p}'\Lambda_{\mathfrak{q}}$ . We obtain  $\varphi : k[y_1, \dots, y_m] \rightarrow \Lambda$  having property (1) such that  $k[y_1, \dots, y_m]_{\mathfrak{p}}/\mathfrak{p}^n k[y_1, \dots, y_m]_{\mathfrak{p}} \rightarrow \Lambda$  factors through  $A$ . Thus  $k[y_1, \dots, y_m]_{\mathfrak{p}}/\mathfrak{p}^n k[y_1, \dots, y_m]_{\mathfrak{p}} \rightarrow A$  is flat by Algebra, Lemma ??. By construction the residue field extension  $F/\kappa(\mathfrak{p})$  is finitely generated and  $\Omega_{F/\kappa(\mathfrak{p})} = 0$ . Hence it is finite separable by More on Algebra, Lemma ??. Thus  $k[y_1, \dots, y_m]_{\mathfrak{p}}/\mathfrak{p}^n k[y_1, \dots, y_m]_{\mathfrak{p}} \rightarrow A$  is finite by Algebra, Lemma ??. Finally, we conclude that it is étale by Algebra, Lemma ??. Since an étale ring map is certainly essentially smooth we win.  $\square$

**Lemma 11.3.** Let  $\varphi : k[y_1, \dots, y_m] \rightarrow \Lambda$ ,  $n$ ,  $\mathfrak{q}$ ,  $\mathfrak{p}$  and

$$k[y_1, \dots, y_m]_{\mathfrak{p}}/\mathfrak{p}^n \rightarrow D \rightarrow \Lambda_{\mathfrak{q}}/\mathfrak{q}^n \Lambda_{\mathfrak{q}}$$

be as in Lemma 11.2. Then for any  $\lambda \in \Lambda \setminus \mathfrak{q}$  there exists an integer  $q > 0$  and a factorization

$$k[y_1, \dots, y_m]_{\mathfrak{p}}/\mathfrak{p}^n \rightarrow D \rightarrow D' \rightarrow \Lambda_{\mathfrak{q}}/\mathfrak{q}^n \Lambda_{\mathfrak{q}}$$

such that  $D \rightarrow D'$  is an essentially smooth map of local Artinian rings, the last arrow is flat, and  $\lambda^q$  is in  $D'$ .

**Proof.** Set  $\bar{\Lambda} = \Lambda_{\mathfrak{q}}/\mathfrak{q}^n \Lambda_{\mathfrak{q}}$ . Let  $\bar{\lambda}$  be the image of  $\lambda$  in  $\bar{\Lambda}$ . Let  $\alpha \in \kappa(\mathfrak{q})$  be the image of  $\lambda$  in the residue field. Let  $k \subset F \subset \kappa(\mathfrak{q})$  be the residue field of  $D$ . If  $\alpha$  is in  $F$  then we can find an  $x \in D$  such that  $x\bar{\lambda} = 1 \bmod \mathfrak{q}$ . Hence  $(x\bar{\lambda})^q = 1 \bmod (\mathfrak{q})^q$  if  $q$  is divisible by  $p$ . Hence  $\bar{\lambda}^q$  is in  $D$ . If  $\alpha$  is transcendental over  $F$ , then we can take  $D' = (D[\bar{\lambda}])_{\mathfrak{m}}$  equal to the subring generated by  $D$  and  $\bar{\lambda}$  localized at  $\mathfrak{m} = D[\bar{\lambda}] \cap \mathfrak{q}\bar{\Lambda}$ . This works because  $D[\bar{\lambda}]$  is in fact a polynomial algebra over  $D$  in this case. Finally, if  $\lambda \bmod \mathfrak{q}$  is algebraic over  $F$ , then we can find a  $p$ -power  $q$  such that  $\alpha^q$  is separable algebraic over  $F$ , see Fields, Section ??. Note that  $D$  and  $\bar{\Lambda}$  are henselian local rings, see Algebra, Lemma ??. Let  $D \rightarrow D'$  be a finite étale extension whose residue field extension is  $F(\alpha^q)/F$ , see Algebra, Lemma ??. Since  $\bar{\Lambda}$  is henselian and  $F(\alpha^q)$  is contained in its residue field we can find a factorization  $D' \rightarrow \bar{\Lambda}$ . By the first part of the argument we see that  $\bar{\lambda}^{qq'} \in D'$  for some  $q' > 0$ .  $\square$

**Lemma 11.4.** Let  $k \rightarrow A \rightarrow \Lambda \supset \mathfrak{q}$  be as in Situation 9.1 where

- (1)  $k$  is a field of characteristic  $p > 0$ ,
- (2)  $\Lambda$  is Noetherian and geometrically regular over  $k$ ,

(3)  $\mathfrak{q}$  is minimal over  $\mathfrak{h}_A$ .

Then  $k \rightarrow A \rightarrow \Lambda \supset \mathfrak{q}$  can be resolved.

**Proof.** The lemma is proven by the following steps in the given order. We will justify each of these steps below.

- (1) Pick an integer  $N > 0$  such that  $\mathfrak{q}^N \Lambda_{\mathfrak{q}} \subset H_{A/k} \Lambda_{\mathfrak{q}}$ .
- (2) Pick generators  $a_1, \dots, a_t \in A$  of the ideal  $H_{A/R}$ .
- (3) Set  $d = \dim(\Lambda_{\mathfrak{q}})$ .
- (4) Set  $B = A[x_1, \dots, x_d, z_{ij}]/(x_i^{2N} - \sum z_{ij} a_j)$ .
- (5) Consider  $B$  as a  $k[x_1, \dots, x_d]$ -algebra and let  $B \rightarrow C$  be as in Lemma 3.1. We also obtain a section  $C \rightarrow B$ .
- (6) Choose  $c > 0$  such that each  $x_i^c$  is strictly standard in  $C$  over  $k[x_1, \dots, x_d]$ .
- (7) Set  $n = N + dc$  and  $e = 8c$ .
- (8) Let  $E \subset \Lambda_{\mathfrak{q}}/\mathfrak{q}^n \Lambda_{\mathfrak{q}}$  be the images of generators of  $A$  as a  $k$ -algebra.
- (9) Choose an integer  $m$  and a  $k$ -algebra map  $\varphi : k[y_1, \dots, y_m] \rightarrow \Lambda$  and a factorization by local Artinian rings

$$k[y_1, \dots, y_m]_{\mathfrak{p}}/\mathfrak{p}^n k[y_1, \dots, y_m]_{\mathfrak{p}} \rightarrow D \rightarrow \Lambda_{\mathfrak{q}}/\mathfrak{q}^n \Lambda_{\mathfrak{q}}$$

such that the first arrow is essentially smooth, the second is flat,  $E$  is contained in  $D$ , with  $\mathfrak{p} = \varphi^{-1}(\mathfrak{q})$  the map  $k[y_1, \dots, y_m]_{\mathfrak{p}} \rightarrow \Lambda_{\mathfrak{q}}$  is flat, and  $\mathfrak{p} \Lambda_{\mathfrak{q}} = \mathfrak{q} \Lambda_{\mathfrak{q}}$ .

- (10) Choose  $\pi_1, \dots, \pi_d \in \mathfrak{p}$  which map to a regular system of parameters of  $k[y_1, \dots, y_m]_{\mathfrak{p}}$ .
- (11) Let  $R = k[y_1, \dots, y_m, t_1, \dots, t_m]$  and  $\gamma_i = \pi_i t_i$ .
- (12) If necessary modify the choice of  $\pi_i$  such that for  $i = 1, \dots, d$  we have

$$\text{Ann}_{R/(\gamma_1^e, \dots, \gamma_{i-1}^e)R}(\gamma_i) = \text{Ann}_{R/(\gamma_1^e, \dots, \gamma_{i-1}^e)R}(\gamma_i^2)$$

- (13) There exist  $\delta_1, \dots, \delta_d \in \Lambda$ ,  $\delta_i \notin \mathfrak{q}$  and a factorization  $D \rightarrow D' \rightarrow \Lambda_{\mathfrak{q}}/\mathfrak{q}^n \Lambda_{\mathfrak{q}}$  with  $D'$  local Artinian,  $D \rightarrow D'$  essentially smooth, the map  $D' \rightarrow \Lambda_{\mathfrak{q}}/\mathfrak{q}^n \Lambda_{\mathfrak{q}}$  flat such that, with  $\pi'_i = \delta_i \pi_i$ , we have for  $i = 1, \dots, d$ 
  - (a)  $(\pi'_i)^{2N} = \sum a_j \lambda_{ij}$  in  $\Lambda$  where  $\lambda_{ij} \bmod \mathfrak{q}^n \Lambda_{\mathfrak{q}}$  is an element of  $D'$ ,
  - (b)  $\text{Ann}_{\Lambda/(\pi_1^e, \dots, \pi_{i-1}^e)}(\pi'_i) = \text{Ann}_{\Lambda/(\pi_1^e, \dots, \pi_{i-1}^e)}(\pi_i'^2)$ ,
  - (c)  $\delta_i \bmod \mathfrak{q}^n \Lambda_{\mathfrak{q}}$  is an element of  $D'$ .
- (14) Define  $B \rightarrow \Lambda$  by sending  $x_i$  to  $\pi'_i$  and  $z_{ij}$  to  $\lambda_{ij}$  found above. Define  $C \rightarrow \Lambda$  by composing the map  $B \rightarrow \Lambda$  with the retraction  $C \rightarrow B$ .
- (15) Map  $R \rightarrow \Lambda$  by  $\varphi$  on  $k[y_1, \dots, y_m]$  and by sending  $t_i$  to  $\delta_i$ . Further introduce a map

$$k[x_1, \dots, x_d] \longrightarrow R = k[y_1, \dots, y_m, t_1, \dots, t_d]$$

by sending  $x_i$  to  $\gamma_i = \pi_i t_i$ .

- (16) It suffices to resolve

$$R \rightarrow C \otimes_{k[x_1, \dots, x_d]} R \rightarrow \Lambda \supset \mathfrak{q}$$

- (17) Set  $I = (\gamma_1^e, \dots, \gamma_d^e) \subset R$ .
- (18) It suffices to resolve

$$R/I \rightarrow C \otimes_{k[x_1, \dots, x_d]} R/I \rightarrow \Lambda/I\Lambda \supset \mathfrak{q}/I\Lambda$$

- (19) We denote  $\mathfrak{r} \subset R = k[y_1, \dots, y_m, t_1, \dots, t_d]$  the inverse image of  $\mathfrak{q}$ .
- (20) It suffices to resolve

$$(R/I)_{\mathfrak{r}} \rightarrow C \otimes_{k[x_1, \dots, x_d]} (R/I)_{\mathfrak{r}} \rightarrow \Lambda_{\mathfrak{q}}/I\Lambda_{\mathfrak{q}} \supset \mathfrak{q}\Lambda_{\mathfrak{q}}/I\Lambda_{\mathfrak{q}}$$

(21) Set  $J = (\pi_1^e, \dots, \pi_d^e)$  in  $k[y_1, \dots, y_m]$ .

(22) It suffices to resolve

$$(R/JR)_{\mathfrak{p}} \rightarrow C \otimes_{k[x_1, \dots, x_d]} (R/JR)_{\mathfrak{p}} \rightarrow \Lambda_{\mathfrak{q}}/J\Lambda_{\mathfrak{q}} \supset \mathfrak{q}\Lambda_{\mathfrak{q}}/J\Lambda_{\mathfrak{q}}$$

(23) It suffices to resolve

$$(R/\mathfrak{p}^n R)_{\mathfrak{p}} \rightarrow C \otimes_{k[x_1, \dots, x_d]} (R/\mathfrak{p}^n R)_{\mathfrak{p}} \rightarrow \Lambda_{\mathfrak{q}}/\mathfrak{q}^n \Lambda_{\mathfrak{q}} \supset \mathfrak{q}\Lambda_{\mathfrak{q}}/\mathfrak{q}^n \Lambda_{\mathfrak{q}}$$

(24) It suffices to resolve

$$(R/\mathfrak{p}^n R)_{\mathfrak{p}} \rightarrow B \otimes_{k[x_1, \dots, x_d]} (R/\mathfrak{p}^n R)_{\mathfrak{p}} \rightarrow \Lambda_{\mathfrak{q}}/\mathfrak{q}^n \Lambda_{\mathfrak{q}} \supset \mathfrak{q}\Lambda_{\mathfrak{q}}/\mathfrak{q}^n \Lambda_{\mathfrak{q}}$$

(25) The ring  $D'[t_1, \dots, t_d]$  is given the structure of an  $R_{\mathfrak{p}}/\mathfrak{p}^n R_{\mathfrak{p}}$ -algebra by the given map  $k[y_1, \dots, y_m]_{\mathfrak{p}}/\mathfrak{p}^n k[y_1, \dots, y_m]_{\mathfrak{p}} \rightarrow D'$  and by sending  $t_i$  to  $t_i$ . It suffices to find a factorization

$$B \otimes_{k[x_1, \dots, x_d]} (R/\mathfrak{p}^n R)_{\mathfrak{p}} \rightarrow D'[t_1, \dots, t_d] \rightarrow \Lambda_{\mathfrak{q}}/\mathfrak{q}^n \Lambda_{\mathfrak{q}}$$

where the second arrow sends  $t_i$  to  $\delta_i$  and induces the given homomorphism  $D' \rightarrow \Lambda_{\mathfrak{q}}/\mathfrak{q}^n \Lambda_{\mathfrak{q}}$ .

(26) Such a factorization exists by our choice of  $D'$  above.

We now give the justification for each of the steps, except that we skip justifying the steps which just introduce notation.

Ad (1). This is possible as  $\mathfrak{q}$  is minimal over  $\mathfrak{h}_A = \sqrt{H_{A/k}\Lambda}$ .

Ad (6). Note that  $A_{a_i}$  is smooth over  $k$ . Hence  $B_{a_j}$ , which is isomorphic to a polynomial algebra over  $A_{a_j}[x_1, \dots, x_d]$ , is smooth over  $k[x_1, \dots, x_d]$ . Thus  $B_{x_i}$  is smooth over  $k[x_1, \dots, x_d]$ . By Lemma 3.1 we see that  $C_{x_i}$  is smooth over  $k[x_1, \dots, x_d]$  with finite free module of differentials. Hence some power of  $x_i$  is strictly standard in  $C$  over  $k[x_1, \dots, x_n]$  by Lemma 3.7.

Ad (9). This follows by applying Lemma 11.2.

Ad (10). Since  $k[y_1, \dots, y_m]_{\mathfrak{p}} \rightarrow \Lambda_{\mathfrak{q}}$  is flat and  $\mathfrak{p}\Lambda_{\mathfrak{q}} = \mathfrak{q}\Lambda_{\mathfrak{q}}$  by construction we see that  $\dim(k[y_1, \dots, y_m]_{\mathfrak{p}}) = d$  by Algebra, Lemma ???. Thus we can find  $\pi_1, \dots, \pi_d \in \Lambda$  which map to a regular system of parameters in  $\Lambda_{\mathfrak{q}}$ .

Ad (12). By Algebra, Lemma ??? any permutation of the sequence  $\pi_1, \dots, \pi_d$  is a regular sequence in  $k[y_1, \dots, y_m]_{\mathfrak{p}}$ . Hence  $\gamma_1 = \pi_1 t_1, \dots, \gamma_d = \pi_d t_d$  is a regular sequence in  $R_{\mathfrak{p}} = k[y_1, \dots, y_m]_{\mathfrak{p}}[t_1, \dots, t_d]$ , see Algebra, Lemma ???. Let  $S = k[y_1, \dots, y_m] \setminus \mathfrak{p}$  so that  $R_{\mathfrak{p}} = S^{-1}R$ . Note that  $\pi_1, \dots, \pi_d$  and  $\gamma_1, \dots, \gamma_d$  remain regular sequences if we multiply our  $\pi_i$  by elements of  $S$ . Suppose that

$$\text{Ann}_{R/(\gamma_1^e, \dots, \gamma_{i-1}^e)R}(\gamma_i) = \text{Ann}_{R/(\gamma_1^e, \dots, \gamma_{i-1}^e)R}(\gamma_i^2)$$

holds for  $i = 1, \dots, t$  for some  $t \in \{0, \dots, d\}$ . Note that  $\gamma_1^e, \dots, \gamma_t^e, \gamma_{t+1}^e$  is a regular sequence in  $S^{-1}R$  by Algebra, Lemma ???. Hence we see that

$$\text{Ann}_{S^{-1}R/(\gamma_1^e, \dots, \gamma_{i-1}^e)R}(\gamma_i) = \text{Ann}_{S^{-1}R/(\gamma_1^e, \dots, \gamma_{i-1}^e)R}(\gamma_i^2).$$

Thus we get

$$\text{Ann}_{R/(\gamma_1^e, \dots, \gamma_t^e)R}(\gamma_{t+1}) = \text{Ann}_{R/(\gamma_1^e, \dots, \gamma_t^e)R}(\gamma_{t+1}^2)$$

after replacing  $\pi_{t+1}$  by  $s\pi_{t+1}$  for some  $s \in S$  by Lemma 10.1. By induction on  $t$  this produces the desired sequence.

Ad (13). Let  $S = \Lambda \setminus \mathfrak{q}$  so that  $\Lambda_{\mathfrak{q}} = S^{-1}\Lambda$ . Set  $\bar{\Lambda} = \Lambda_{\mathfrak{q}}/\mathfrak{q}^n \Lambda_{\mathfrak{q}}$ . Suppose that we have a  $t \in \{0, \dots, d\}$  and  $\delta_1, \dots, \delta_t \in S$  and a factorization  $D \rightarrow D' \rightarrow \bar{\Lambda}$  as

in (13) such that (a), (b), (c) hold for  $i = 1, \dots, t$ . We have  $\pi_{t+1}^N \in H_{A/k}\Lambda_{\mathfrak{q}}$  as  $\mathfrak{q}^N\Lambda_{\mathfrak{q}} \subset H_{A/k}\Lambda_{\mathfrak{q}}$  by (1). Hence  $\pi_{t+1}^N \in H_{A/k}\bar{\Lambda}$ . Hence  $\pi_{t+1}^N \in H_{A/k}D'$  as  $D' \rightarrow \bar{\Lambda}$  is faithfully flat, see Algebra, Lemma ???. Recall that  $H_{A/k} = (a_1, \dots, a_t)$ . Say  $\pi_{t+1}^N = \sum a_j d_j$  in  $D'$  and choose  $c_j \in \Lambda_{\mathfrak{q}}$  lifting  $d_j \in D'$ . Then  $\pi_{t+1}^N = \sum c_j a_j + \epsilon$  with  $\epsilon \in \mathfrak{q}^N\Lambda_{\mathfrak{q}} \subset \mathfrak{q}^{n-N}H_{A/k}\Lambda_{\mathfrak{q}}$ . Write  $\epsilon = \sum a_j c'_j$  for some  $c'_j \in \mathfrak{q}^{n-N}\Lambda_{\mathfrak{q}}$ . Hence  $\pi_{t+1}^{2N} = \sum (\pi_{t+1}^N c_j + \pi_{t+1}^N c'_j) a_j$ . Note that  $\pi_{t+1}^N c'_j$  maps to zero in  $\bar{\Lambda}$ ; this trivial but key observation will ensure later that (a) holds. Now we choose  $s \in S$  such that there exist  $\mu_{t+1j} \in \Lambda$  such that on the one hand  $\pi_{t+1}^N c_j + \pi_{t+1}^N c'_j = \mu_{t+1j}/s^{2N}$  in  $S^{-1}\Lambda$  and on the other  $(s\pi_{t+1})^{2N} = \sum \mu_{t+1j} a_j$  in  $\Lambda$  (minor detail omitted). We may further replace  $s$  by a power and enlarge  $D'$  such that  $s$  maps to an element of  $D'$ . With these choices  $\mu_{t+1j}$  maps to  $s^{2N} d_j$  which is an element of  $D'$ . Note that  $\pi_1, \dots, \pi_d$  are a regular sequence of parameters in  $S^{-1}\Lambda$  by our choice of  $\varphi$ . Hence  $\pi_1, \dots, \pi_d$  forms a regular sequence in  $\Lambda_{\mathfrak{q}}$  by Algebra, Lemma ???. It follows that  $\pi_1^e, \dots, \pi_t^e, s\pi_{t+1}$  is a regular sequence in  $S^{-1}\Lambda$  by Algebra, Lemma ???. Thus we get

$$\text{Ann}_{S^{-1}\Lambda/(\pi_1^e, \dots, \pi_t^e)}(s\pi_{t+1}) = \text{Ann}_{S^{-1}\Lambda/(\pi_1^e, \dots, \pi_t^e)}((s\pi_{t+1})^2).$$

Hence we may apply Lemma 10.1 to find an  $s' \in S$  such that

$$\text{Ann}_{\Lambda/(\pi_1^e, \dots, \pi_t^e)}((s')^q s\pi_{t+1}) = \text{Ann}_{\Lambda/(\pi_1^e, \dots, \pi_t^e)}(((s')^q s\pi_{t+1})^2).$$

for any  $q > 0$ . By Lemma 11.3 we can choose  $q$  and enlarge  $D'$  such that  $(s')^q$  maps to an element of  $D'$ . Setting  $\delta_{t+1} = (s')^q s$  and we conclude that (a), (b), (c) hold for  $i = 1, \dots, t+1$ . For (a) note that  $\lambda_{t+1j} = (s')^{2Nq} \mu_{t+1j}$  works. By induction on  $t$  we win.

Ad (16). By construction the radical of  $H_{(C \otimes_{k[x_1, \dots, x_d]} R)/R} \Lambda$  contains  $\mathfrak{h}_A$ . Namely, the elements  $a_j \in H_{A/k}$  map to elements of  $H_{B/k[x_1, \dots, x_n]}$ , hence map to elements of  $H_{C/k[x_1, \dots, x_n]}$ , hence  $a_j \otimes 1$  map to elements of  $H_{C \otimes_{k[x_1, \dots, x_d]} R/R}$ . Moreover, if we have a solution  $C \otimes_{k[x_1, \dots, x_n]} R \rightarrow T \rightarrow \Lambda$  of

$$R \rightarrow C \otimes_{k[x_1, \dots, x_d]} R \rightarrow \Lambda \supset \mathfrak{q}$$

then  $H_{T/R} \subset H_{T/k}$  as  $R$  is smooth over  $k$ . Hence  $T$  will also be a solution for the original situation  $k \rightarrow A \rightarrow \Lambda \supset \mathfrak{q}$ .

Ad (18). Follows on applying Lemma 9.2 to  $R \rightarrow C \otimes_{k[x_1, \dots, x_d]} R \rightarrow \Lambda \supset \mathfrak{q}$  and the sequence of elements  $\gamma_1^c, \dots, \gamma_d^c$ . We note that since  $x_i^c$  are strictly standard in  $C$  over  $k[x_1, \dots, x_d]$  the elements  $\gamma_i^c$  are strictly standard in  $C \otimes_{k[x_1, \dots, x_d]} R$  over  $R$  by Lemma 2.7. The other assumption of Lemma 9.2 holds by steps (12) and (13).

Ad (20). Apply Lemma 9.4 to the situation in (18). In the rest of the arguments the target ring is local Artinian, hence we are looking for a factorization by a smooth algebra  $T$  over the source ring.

Ad (22). Suppose that  $C \otimes_{k[x_1, \dots, x_d]} (R/JR)_{\mathfrak{p}} \rightarrow T \rightarrow \Lambda_{\mathfrak{q}}/J\Lambda_{\mathfrak{q}}$  is a solution to

$$(R/JR)_{\mathfrak{p}} \rightarrow C \otimes_{k[x_1, \dots, x_d]} (R/JR)_{\mathfrak{p}} \rightarrow \Lambda_{\mathfrak{q}}/J\Lambda_{\mathfrak{q}} \supset \mathfrak{q}\Lambda_{\mathfrak{q}}/J\Lambda_{\mathfrak{q}}$$

Then  $C \otimes_{k[x_1, \dots, x_d]} (R/I)_{\mathfrak{r}} \rightarrow T_{\mathfrak{r}} \rightarrow \Lambda_{\mathfrak{q}}/I\Lambda_{\mathfrak{q}}$  is a solution to the situation in (20).

Ad (23). Our  $n = N + dc$  is large enough so that  $\mathfrak{p}^n k[y_1, \dots, y_m]_{\mathfrak{p}} \subset J_{\mathfrak{p}}$  and  $\mathfrak{q}^n \Lambda_{\mathfrak{q}} \subset J\Lambda_{\mathfrak{q}}$ . Hence if we have a solution  $C \otimes_{k[x_1, \dots, x_d]} (R/\mathfrak{p}^n R)_{\mathfrak{p}} \rightarrow T \rightarrow \Lambda_{\mathfrak{q}}/\mathfrak{q}^n \Lambda_{\mathfrak{q}}$  of (22) then we can take  $T/JT$  as the solution for (23).

Ad (24). This is true because we have a section  $C \rightarrow B$  in the category of  $R$ -algebras.

Ad (25). This is true because  $D'$  is essentially smooth over the local Artinian ring  $k[y_1, \dots, y_m]_{\mathfrak{p}}/\mathfrak{p}^n k[y_1, \dots, y_m]_{\mathfrak{p}}$  and

$$R_{\mathfrak{p}}/\mathfrak{p}^n R_{\mathfrak{p}} = k[y_1, \dots, y_m]_{\mathfrak{p}}/\mathfrak{p}^n k[y_1, \dots, y_m]_{\mathfrak{p}}[t_1, \dots, t_d].$$

Hence  $D'[t_1, \dots, t_d]$  is a filtered colimit of smooth  $R_{\mathfrak{p}}/\mathfrak{p}^n R_{\mathfrak{p}}$ -algebras and  $B \otimes_{k[x_1, \dots, x_d]} (R_{\mathfrak{p}}/\mathfrak{p}^n R_{\mathfrak{p}})$  factors through one of these.

Ad (26). The final twist of the proof is that we cannot just use the map  $B \rightarrow D'$  which maps  $x_i$  to the image of  $\pi'_i$  in  $D'$  and  $z_{ij}$  to the image of  $\lambda_{ij}$  in  $D'$  because we need the diagram

$$\begin{array}{ccc} B & \longrightarrow & D'[t_1, \dots, t_d] \\ \uparrow & & \uparrow \\ k[x_1, \dots, x_d] & \longrightarrow & R_{\mathfrak{p}}/\mathfrak{p}^n R_{\mathfrak{p}} \end{array}$$

to commute and we need the composition  $B \rightarrow D'[t_1, \dots, t_d] \rightarrow \Lambda_{\mathfrak{q}}/\mathfrak{q}^n \Lambda_{\mathfrak{q}}$  to be the map of (14). This requires us to map  $x_i$  to the image of  $\pi_i t_i$  in  $D'[t_1, \dots, t_d]$ . Hence we map  $z_{ij}$  to the image of  $\lambda_{ij} t_i^{2N}/\delta_i^{2N}$  in  $D'[t_1, \dots, t_d]$  and everything is clear.  $\square$

## 12. The main theorem

In this section we wrap up the discussion.

**Theorem 12.1** (Popescu). *Any regular homomorphism of Noetherian rings is a filtered colimit of smooth ring maps.*

**Proof.** By Lemma 8.4 it suffices to prove this for  $k \rightarrow \Lambda$  where  $\Lambda$  is Noetherian and geometrically regular over  $k$ . Let  $k \rightarrow A \rightarrow \Lambda$  be a factorization with  $A$  a finite type  $k$ -algebra. It suffices to construct a factorization  $A \rightarrow B \rightarrow \Lambda$  with  $B$  of finite type such that  $\mathfrak{h}_B = \Lambda$ , see Lemma 2.8. Hence we may perform Noetherian induction on the ideal  $\mathfrak{h}_A$ . Pick a prime  $\mathfrak{q} \supset \mathfrak{h}_A$  such that  $\mathfrak{q}$  is minimal over  $\mathfrak{h}_A$ . It now suffices to resolve  $k \rightarrow A \rightarrow \Lambda \supset \mathfrak{q}$  (as defined in the text following Situation 9.1). If the characteristic of  $k$  is zero, this follows from Lemma 10.3. If the characteristic of  $k$  is  $p > 0$ , this follows from Lemma 11.4.  $\square$

## 13. The approximation property for G-rings

Let  $R$  be a Noetherian local ring. In this case  $R$  is a G-ring if and only if the ring map  $R \rightarrow R^\wedge$  is regular, see More on Algebra, Lemma ???. In this case it is true that the henselization  $R^h$  and the strict henselization  $R^{sh}$  of  $R$  are G-rings, see More on Algebra, Lemma ???. Moreover, any algebra essentially of finite type over a field, over a complete local ring, over  $\mathbf{Z}$ , or over a characteristic zero Dedekind ring is a G-ring, see More on Algebra, Proposition ???. This gives an ample supply of rings to which the result below applies.

Let  $R$  be a ring. Let  $f_1, \dots, f_m \in R[x_1, \dots, x_n]$ . Let  $S$  be an  $R$ -algebra. In this situation we say a vector  $(a_1, \dots, a_n) \in S^n$  is a *solution in  $S$*  if and only if

$$f_j(a_1, \dots, a_n) = 0 \text{ in } S, \text{ for } j = 1, \dots, m$$

Of course an important question in algebraic geometry is to see when systems of polynomial equations have solutions. The following theorem tells us that having solutions in the completion of a local Noetherian ring is often enough to show there exist solutions in the henselization of the ring.

**Theorem 13.1.** *Let  $R$  be a Noetherian local ring. Let  $f_1, \dots, f_m \in R[x_1, \dots, x_n]$ . Suppose that  $(a_1, \dots, a_n) \in (R^\wedge)^n$  is a solution in  $R^\wedge$ . If  $R$  is a henselian G-ring, then for every integer  $N$  there exists a solution  $(b_1, \dots, b_n) \in R^n$  in  $R$  such that  $a_i - b_i \in \mathfrak{m}^N R^\wedge$ .*

**Proof.** Let  $c_i \in R$  be an element such that  $a_i - c_i \in \mathfrak{m}^N$ . Choose generators  $\mathfrak{m}^N = (d_1, \dots, d_M)$ . Write  $a_i = c_i + \sum a_{i,l} d_l$ . Consider the polynomial ring  $R[x_{i,l}]$  and the elements

$$g_j = f_j(c_1 + \sum x_{1,l} d_l, \dots, c_n + \sum x_{n,l} d_{n,l}) \in R[x_{i,l}]$$

The system of equations  $g_j = 0$  has the solution  $(a_{i,l})$ . Suppose that we can show that  $g_j$  as a solution  $(b_{i,l})$  in  $R$ . Then it follows that  $b_i = c_i + \sum b_{i,l} d_l$  is a solution of  $f_j = 0$  which is congruent to  $a_i$  modulo  $\mathfrak{m}^N$ . Thus it suffices to show that solvability over  $R^\wedge$  implies solvability over  $R$ .

Let  $A \subset R^\wedge$  be the  $R$ -subalgebra generated by  $a_1, \dots, a_n$ . Since we've assumed  $R$  is a G-ring, i.e., that  $R \rightarrow R^\wedge$  is regular, we see that there exists a factorization

$$A \rightarrow B \rightarrow R^\wedge$$

with  $B$  smooth over  $R$ , see Theorem 12.1. Denote  $\kappa = R/\mathfrak{m}$  the residue field. It is also the residue field of  $R^\wedge$ , so we get a commutative diagram

$$\begin{array}{ccc} B & \xrightarrow{\quad} & R' \\ \uparrow & \searrow & \downarrow \\ R & \xrightarrow{\quad} & \kappa \end{array}$$

Since the vertical arrow is smooth, More on Algebra, Lemma ?? implies that there exists an étale ring map  $R \rightarrow R'$  which induces an isomorphism  $R/\mathfrak{m} \rightarrow R'/\mathfrak{m}R'$  and an  $R$ -algebra map  $B \rightarrow R'$  making the diagram above commute. Since  $R$  is henselian we see that  $R \rightarrow R'$  has a section, see Algebra, Lemma ??. Let  $b_i \in R$  be the image of  $a_i$  under the ring maps  $A \rightarrow B \rightarrow R' \rightarrow R$ . Since all of these maps are  $R$ -algebra maps, we see that  $(b_1, \dots, b_n)$  is a solution in  $R$ .  $\square$

Given a Noetherian local ring  $(R, \mathfrak{m})$ , an étale ring map  $R \rightarrow R'$ , and a maximal ideal  $\mathfrak{m}' \subset R'$  lying over  $\mathfrak{m}$  with  $\kappa(\mathfrak{m}) = \kappa(\mathfrak{m}')$ , then we have inclusions

$$R \subset R_{\mathfrak{m}'} \subset R^h \subset R^\wedge,$$

by Algebra, Lemma ?? and More on Algebra, Lemma ??.

**Theorem 13.2.** *Let  $R$  be a Noetherian local ring. Let  $f_1, \dots, f_m \in R[x_1, \dots, x_n]$ . Suppose that  $(a_1, \dots, a_n) \in (R^\wedge)^n$  is a solution. If  $R$  is a G-ring, then for every integer  $N$  there exist*

- (1) an étale ring map  $R \rightarrow R'$ ,
- (2) a maximal ideal  $\mathfrak{m}' \subset R'$  lying over  $\mathfrak{m}$
- (3) a solution  $(b_1, \dots, b_n) \in (R')^n$  in  $R'$

such that  $\kappa(\mathfrak{m}) = \kappa(\mathfrak{m}')$  and  $a_i - b_i \in (\mathfrak{m}')^N R^\wedge$ .



**Proof.** We could deduce this theorem from Theorem 13.1 using that the henselization  $R^h$  is a G-ring by More on Algebra, Lemma ?? and writing  $R^h$  as a directed colimit of étale extension  $R'$ . Instead we prove this by redoing the proof of the previous theorem in this case.

Let  $c_i \in R$  be an element such that  $a_i - c_i \in \mathfrak{m}^N$ . Choose generators  $\mathfrak{m}^N = (d_1, \dots, d_M)$ . Write  $a_i = c_i + \sum a_{i,l} d_l$ . Consider the polynomial ring  $R[x_{i,l}]$  and the elements

$$g_j = f_j(c_1 + \sum x_{1,l} d_l, \dots, c_n + \sum x_{n,l} d_{n,l}) \in R[x_{i,l}]$$

The system of equations  $g_j = 0$  has the solution  $(a_{i,l})$ . Suppose that we can show that  $g_j$  as a solution  $(b_{i,l})$  in  $R'$  for some étale ring map  $R \rightarrow R'$  endowed with a maximal ideal  $\mathfrak{m}'$  such that  $\kappa(\mathfrak{m}) = \kappa(\mathfrak{m}')$ . Then it follows that  $b_i = c_i + \sum b_{i,l} d_l$  is a solution of  $f_j = 0$  which is congruent to  $a_i$  modulo  $(\mathfrak{m}')^N$ . Thus it suffices to show that solvability over  $R^\wedge$  implies solvability over some étale ring extension which induces a trivial residue field extension at some prime over  $\mathfrak{m}$ .

Let  $A \subset R^\wedge$  be the  $R$ -subalgebra generated by  $a_1, \dots, a_n$ . Since we've assumed  $R$  is a G-ring, i.e., that  $R \rightarrow R^\wedge$  is regular, we see that there exists a factorization

$$A \rightarrow B \rightarrow R^\wedge$$

with  $B$  smooth over  $R$ , see Theorem 12.1. Denote  $\kappa = R/\mathfrak{m}$  the residue field. It is also the residue field of  $R^\wedge$ , so we get a commutative diagram

$$\begin{array}{ccc} B & \xrightarrow{\quad} & R' \\ \uparrow & \searrow & \downarrow \\ R & \xrightarrow{\quad} & \kappa \end{array}$$

Since the vertical arrow is smooth, More on Algebra, Lemma ?? implies that there exists an étale ring map  $R \rightarrow R'$  which induces an isomorphism  $R/\mathfrak{m} \rightarrow R'/\mathfrak{m}R'$  and an  $R$ -algebra map  $B \rightarrow R'$  making the diagram above commute. Let  $b_i \in R'$  be the image of  $a_i$  under the ring maps  $A \rightarrow B \rightarrow R'$ . Since all of these maps are  $R$ -algebra maps, we see that  $(b_1, \dots, b_n)$  is a solution in  $R'$ .  $\square$

**Example 13.3.** Let  $(R, \mathfrak{m})$  be a Noetherian local ring with henselization  $R^h$ . The map on completions  $R^\wedge \rightarrow (R^h)^\wedge$  is an isomorphism, see More on Algebra, Lemma ??. Since also  $R^h$  is Noetherian (ibid.) we may think of  $R^h$  as a subring of its completion (because the completion is faithfully flat). In this way we see that we may identify  $R^h$  with a subring of  $R^\wedge$ .

Let us try to understand which elements of  $R^\wedge$  are in  $R^h$ . For simplicity we assume  $R$  is a domain with fraction field  $K$ . Clearly, every element  $f$  of  $R^h$  is algebraic over  $R$ , in the sense that there exists an equation of the form  $a_n f^n + \dots + a_1 f + a_0 = 0$  for some  $a_i \in R$  with  $n > 0$  and  $a_n \neq 0$ .

Conversely, assume that  $f \in R^\wedge$ ,  $n \in \mathbb{N}$ , and  $a_0, \dots, a_n \in R$  with  $a_n \neq 0$  such that  $a_n f^n + \dots + a_1 f + a_0 = 0$ . If  $R$  is a G-ring, then, for every  $N > 0$  there exists an element  $g \in R^h$  with  $a_n g^n + \dots + a_1 g + a_0 = 0$  and  $f - g \in \mathfrak{m}^N R^\wedge$ , see Theorem 13.2. We'd like to conclude that  $f = g$  when  $N \gg 0$ . If this is not true, then we find infinitely many roots  $g$  of  $P(T)$  in  $R^h$ . This is impossible because (1)  $R^h \subset R^h \otimes_R K$  and (2)  $R^h \otimes_R K$  is a finite product of field extensions of  $K$ .

Namely,  $R \rightarrow K$  is injective and  $R \rightarrow R^h$  is flat, hence  $R^h \rightarrow R^h \otimes_R K$  is injective and (2) follows from More on Algebra, Lemma ??.

Conclusion: If  $R$  is a Noetherian local domain with fraction field  $K$  and a  $G$ -ring, then  $R^h \subset R^\wedge$  is the set of all elements which are algebraic over  $K$ .

Here is another variant of the main theorem of this section.

**Lemma 13.4.** *Let  $R$  be a Noetherian ring. Let  $\mathfrak{p} \subset R$  be a prime ideal. Let  $f_1, \dots, f_m \in R[x_1, \dots, x_n]$ . Suppose that  $(a_1, \dots, a_n) \in ((R_{\mathfrak{p}})^\wedge)^n$  is a solution. If  $R_{\mathfrak{p}}$  is a  $G$ -ring, then for every integer  $N$  there exist*

- (1) *an étale ring map  $R \rightarrow R'$ ,*
- (2) *a prime ideal  $\mathfrak{p}' \subset R'$  lying over  $\mathfrak{p}$*
- (3) *a solution  $(b_1, \dots, b_n) \in (R')^n$  in  $R'$*

*such that  $\kappa(\mathfrak{p}) = \kappa(\mathfrak{p}')$  and  $a_i - b_i \in (\mathfrak{p}')^N (R'_{\mathfrak{p}'})^\wedge$ .*

**Proof.** By Theorem 13.2 we can find a solution  $(b'_1, \dots, b'_n)$  in some ring  $R''$  étale over  $R_{\mathfrak{p}}$  which comes with a prime ideal  $\mathfrak{p}''$  lying over  $\mathfrak{p}$  such that  $\kappa(\mathfrak{p}) = \kappa(\mathfrak{p}'')$  and  $a_i - b'_i \in (\mathfrak{p}'')^N (R''_{\mathfrak{p}''})^\wedge$ . We can write  $R'' = R' \otimes_R R_{\mathfrak{p}}$  for some étale  $R$ -algebra  $R'$  (see Algebra, Lemma ??). After replacing  $R'$  by a principal localization if necessary we may assume  $(b'_1, \dots, b'_n)$  come from a solution  $(b_1, \dots, b_n)$  in  $R'$ . Setting  $\mathfrak{p}' = R' \cap \mathfrak{p}''$  we see that  $R''_{\mathfrak{p}''} = R'_{\mathfrak{p}'}$ , which finishes the proof.  $\square$

#### 14. Approximation for henselian pairs

We can generalize the discussion of Section 13 to the case of henselian pairs. Henselian pairs were defined in More on Algebra, Section ??.

**Lemma 14.1.** *Let  $(A, I)$  be a henselian pair with  $A$  Noetherian. Let  $A^\wedge$  be the  $I$ -adic completion of  $A$ . Assume at least one of the following conditions holds*

- (1)  *$A \rightarrow A^\wedge$  is a regular ring map,*
- (2)  *$A$  is a Noetherian  $G$ -ring, or*
- (3)  *$(A, I)$  is the henselization (More on Algebra, Lemma ??) of a pair  $(B, J)$  where  $B$  is a Noetherian  $G$ -ring.*

*Given  $f_1, \dots, f_m \in A[x_1, \dots, x_n]$  and  $\hat{a}_1, \dots, \hat{a}_n \in A^\wedge$  such that  $f_j(\hat{a}_1, \dots, \hat{a}_n) = 0$  for  $j = 1, \dots, m$ , for every  $N \geq 1$  there exist  $a_1, \dots, a_n \in A$  such that  $\hat{a}_i - a_i \in I^N$  and such that  $f_j(a_1, \dots, a_n) = 0$  for  $j = 1, \dots, m$ .*

**Proof.** By More on Algebra, Lemma ?? we see that (3) implies (2). By More on Algebra, Lemma ?? we see that (2) implies (1). Thus it suffices to prove the lemma in case  $A \rightarrow A^\wedge$  is a regular ring map.

Let  $\hat{a}_1, \dots, \hat{a}_n$  be as in the statement of the lemma. By Theorem 12.1 we can find a factorization  $A \rightarrow B \rightarrow A^\wedge$  with  $A \rightarrow B$  smooth and  $b_1, \dots, b_n \in B$  with  $f_j(b_1, \dots, b_n) = 0$  in  $B$ . Denote  $\sigma : B \rightarrow A^\wedge \rightarrow A/I^N$  the composition. By More on Algebra, Lemma ?? we can find an étale ring map  $A \rightarrow A'$  which induces an isomorphism  $A/I^N \rightarrow A'/I^N A'$  and an  $A$ -algebra map  $\tilde{\sigma} : B \rightarrow A'$  lifting  $\sigma$ . Since  $(A, I)$  is henselian, there is an  $A$ -algebra map  $\chi : A' \rightarrow A$ , see More on Algebra, Lemma ??. Then setting  $a_i = \chi(\tilde{\sigma}(b_i))$  gives a solution.  $\square$

## 15. Other chapters

### Preliminaries

- (1) Introduction
- (2) Conventions
- (3) Set Theory
- (4) Categories
- (5) Topology
- (6) Sheaves on Spaces
- (7) Sites and Sheaves
- (8) Stacks
- (9) Fields
- (10) Commutative Algebra
- (11) Brauer Groups
- (12) Homological Algebra
- (13) Derived Categories
- (14) Simplicial Methods
- (15) More on Algebra
- (16) Smoothing Ring Maps
- (17) Sheaves of Modules
- (18) Modules on Sites
- (19) Injectives
- (20) Cohomology of Sheaves
- (21) Cohomology on Sites
- (22) Differential Graded Algebra
- (23) Divided Power Algebra
- (24) Differential Graded Sheaves
- (25) Hypercoverings

### Schemes

- (26) Schemes
- (27) Constructions of Schemes
- (28) Properties of Schemes
- (29) Morphisms of Schemes
- (30) Cohomology of Schemes
- (31) Divisors
- (32) Limits of Schemes
- (33) Varieties
- (34) Topologies on Schemes
- (35) Descent
- (36) Derived Categories of Schemes
- (37) More on Morphisms
- (38) More on Flatness
- (39) Groupoid Schemes
- (40) More on Groupoid Schemes
- (41) Étale Morphisms of Schemes

### Topics in Scheme Theory

- (42) Chow Homology

- (43) Intersection Theory
- (44) Picard Schemes of Curves
- (45) Weil Cohomology Theories
- (46) Adequate Modules
- (47) Dualizing Complexes
- (48) Duality for Schemes
- (49) Discriminants and Differents
- (50) de Rham Cohomology
- (51) Local Cohomology
- (52) Algebraic and Formal Geometry
- (53) Algebraic Curves
- (54) Resolution of Surfaces
- (55) Semistable Reduction
- (56) Functors and Morphisms
- (57) Derived Categories of Varieties
- (58) Fundamental Groups of Schemes
- (59) Étale Cohomology
- (60) Crystalline Cohomology
- (61) Pro-étale Cohomology
- (62) Relative Cycles
- (63) More Étale Cohomology
- (64) The Trace Formula

### Algebraic Spaces

- (65) Algebraic Spaces
- (66) Properties of Algebraic Spaces
- (67) Morphisms of Algebraic Spaces
- (68) Decent Algebraic Spaces
- (69) Cohomology of Algebraic Spaces
- (70) Limits of Algebraic Spaces
- (71) Divisors on Algebraic Spaces
- (72) Algebraic Spaces over Fields
- (73) Topologies on Algebraic Spaces
- (74) Descent and Algebraic Spaces
- (75) Derived Categories of Spaces
- (76) More on Morphisms of Spaces
- (77) Flatness on Algebraic Spaces
- (78) Groupoids in Algebraic Spaces
- (79) More on Groupoids in Spaces
- (80) Bootstrap
- (81) Pushouts of Algebraic Spaces

### Topics in Geometry

- (82) Chow Groups of Spaces
- (83) Quotients of Groupoids
- (84) More on Cohomology of Spaces
- (85) Simplicial Spaces
- (86) Duality for Spaces

|                                       |                                      |
|---------------------------------------|--------------------------------------|
| (87) Formal Algebraic Spaces          | (103) Cohomology of Algebraic Stacks |
| (88) Algebraization of Formal Spaces  | (104) Derived Categories of Stacks   |
| (89) Resolution of Surfaces Revisited | (105) Introducing Algebraic Stacks   |
| Deformation Theory                    | (106) More on Morphisms of Stacks    |
| (90) Formal Deformation Theory        | (107) The Geometry of Stacks         |
| (91) Deformation Theory               | Topics in Moduli Theory              |
| (92) The Cotangent Complex            | (108) Moduli Stacks                  |
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| Algebraic Stacks                      | Miscellany                           |
| (94) Algebraic Stacks                 | (110) Examples                       |
| (95) Examples of Stacks               | (111) Exercises                      |
| (96) Sheaves on Algebraic Stacks      | (112) Guide to Literature            |
| (97) Criteria for Representability    | (113) Desirables                     |
| (98) Artin's Axioms                   | (114) Coding Style                   |
| (99) Quot and Hilbert Spaces          | (115) Obsolete                       |
| (100) Properties of Algebraic Stacks  | (116) GNU Free Documentation License |
| (101) Morphisms of Algebraic Stacks   | (117) Auto Generated Index           |
| (102) Limits of Algebraic Stacks      |                                      |

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